

THE TRACE

Pandemic preparedness has a data problem

Nguyen Minh Phat

THE TRACE

Pandemic preparedness has a data problem

First edition, 2026. A monograph.

Written by Nguyen Minh Phat, laboratory specialist, Vietnam.

This book is a proposal. It is not a report of completed work, and it is not clinical, legal or financial advice. Where the evidence is thin, the text says so. Where a claim has never been tested, the text says that too. Readers are asked to check the sources rather than to take the argument on trust; that is the whole point of it.

Every factual claim carries a citation to a named source. Passages drawn from the author's own working experience are marked as such and carry no citation, because they are testimony rather than evidence.

Set in Instrument Serif and Geist. Cover and figures generated from source code; nothing in this book was traced from an image.

Contents

What Nobody Sees 5

Timekeeping or Scorekeeping 23

Three Readers, One Record 42

From Record to Door..... 59

Plugging Into Rails That Already Exist 74

What Could Go Wrong 91

A Small Pilot, and an Invitation 108

What Nobody Sees

I spent five years as a laboratory specialist inside a provincial CDC in Vietnam. I am writing this proposal now because I lived through these events myself. I do not write as an academic who studied these systems from a distance. I write from the physical memory of my own hands. To understand the proposal I am presenting, we must look closely at how the samples arrived in the second half of 2021.

The arrival of these specimens was the first critical step of our daily labor. The samples were collected in the field by health workers. Sometimes, I went out into the field to collect these samples myself. Other times, the district health centers gathered the specimens and sent them back to our central provincial facility.

Every single batch of these samples arrived at our laboratory inside cold boxes. We did not have specialized or uniform medical transport equipment. Instead, we had to rely on whatever cold boxes were available to the units at that moment. These cold boxes were transported to us by whatever means possible. Some came in an ambulance if the unit was fortunate enough to have one. If an ambulance was not available, the units carried the boxes in any vehicle they could find.

The specimens travelled in standard triple packaging: tubes of transport medium inside sealed bags, inside larger bags, inside the cold box.

This specific packaging protocol was passed down through a very clear chain of training. The Pasteur Institute first taught the protocol to the provincial CDC. The CDC then passed this training down to the frontline workers. From there, the frontline workers showed each other how to perform the collection and the packing.

The samples we received came from many different sources. Some were collected from quarantine facilities housing citizens who had returned from abroad. Others came from deep inside the local prisons. Some samples even came from the dead. Each of these different sources sent their specimens to our laboratory in the same types of packages.

Eventually, the sheer quantity of these incoming shipments became too much for our testing facility to hold. When the rising sample volume completely overwhelmed our laboratory space, we had no choice but to move the containers. The incoming cold boxes had to go out into the wide corridor. They lined the hallway in long rows.

This physical arrival of cold boxes, nested bags, and container tubes was our daily reality. By looking at how these samples arrived, we see the foundation of our work and the heavy responsibility of maintaining the trace. This is the background for the system I propose today.

The dangerous part of a pandemic is easy to picture: protective suits, negative pressure, live virus. That part was real. It was not where we lost the most.

The deepest bottleneck of the entire pandemic did not occur in a high-security bio-isolation unit. It was labelling, and labelling is not what the word suggests. The name makes it sound like fieldwork, something done in the heat of the moment under layers of gear.

The truth is much more mundane. We did the labelling before anyone ever went out to collect a single sample. We sat at wooden desks. We wore our everyday, ordinary clothes. No protective gowns were needed because there were no active samples in the room yet. The virus was not present. There was only a stack of blank sheets, a printer, and thousands of empty, clean container tubes.

These tubes contained only the transport medium. The medium was a sterile, red liquid waiting for a swab. Before those tubes could be sent out to hospitals, clinics, or quarantine camps, they had to be marked. Each tube needed a specific identity. So we sat there in our civilian clothes, matching lists on a computer screen to physical labels. We stuck the printed paper onto the cold plastic of the empty tubes.

It was silent, repetitive work. It was administrative. Yet, if a single mistake was made during this quiet desk task, the entire scientific apparatus downstream fell apart. A typo in a birth year or a misaligned barcode on an empty tube broke the link between a person and their tube. Some of those errors were caught later, when a clinician noticed that a name and a date of birth did not agree and sent the query back to us. Catching them cost time we did not have, and the ones nobody noticed became a person waiting at home for a result that could not be matched to them. The danger of a broken trace did not start with a sneeze or a cough. It started with a pen or a keyboard in a quiet office.

We were fighting a war of logistics with the tools of an office clerk.

By understanding that our greatest barrier was a paperwork bottleneck, we can see the true value of the solution I am proposing. Containment infrastructure does its own job and nothing here replaces it. This is a different gap. It sits upstream of the bench, and

closing it costs far less. Fix the trace where it begins: secure the information on the desk before the tube ever meets the patient.

Today, as I write down this proposal, I want to describe the exact workflow we built because it holds the core of my entire idea.

Our daily work always began with a list of patient names. This collection list did not arrive in a formal or highly structured digital format. It usually came to us through a simple email or a common messaging app. We would open the message and print the raw list onto sheets of paper.

Then the slow, manual work began. I sat at my wooden desk and converted that printed paper list by hand into a digital Excel file. This was not a sophisticated program. It was just an ordinary spreadsheet, but we used it as our central database to organize the chaos.

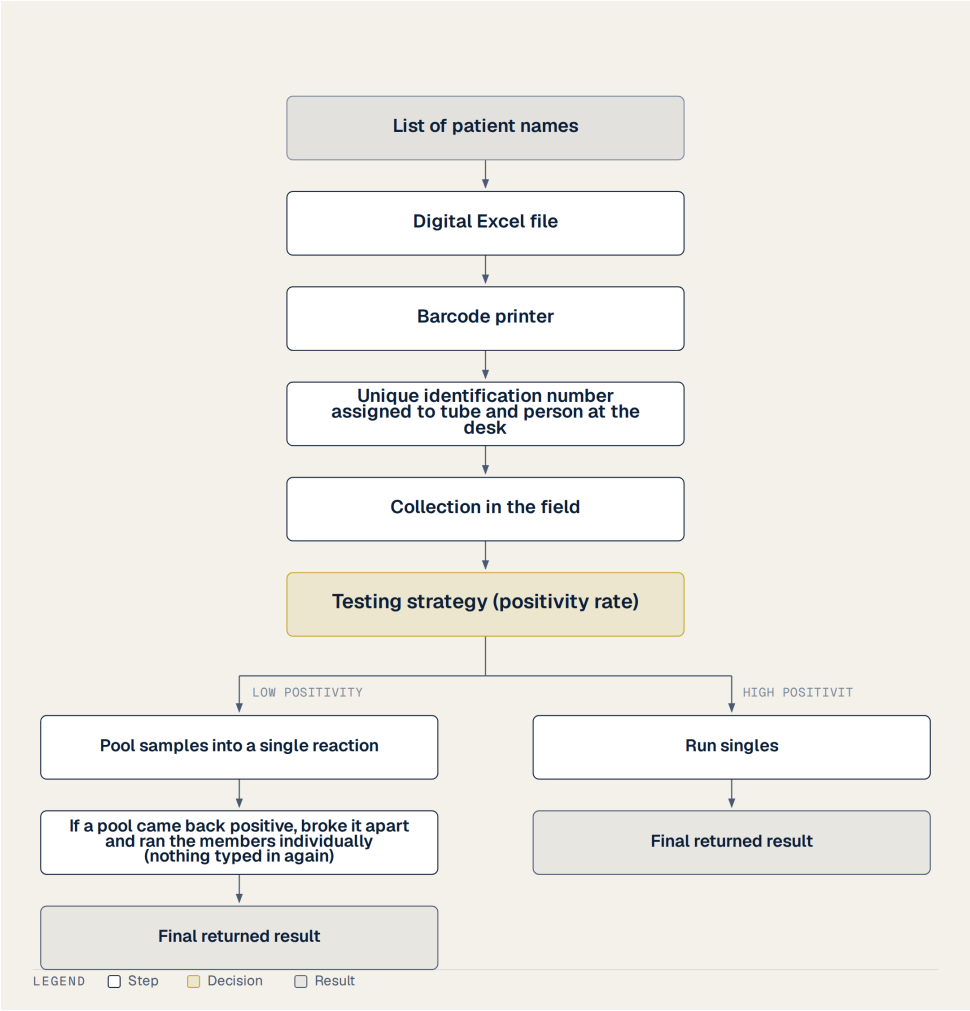
We linked this Excel database directly to a barcode printer. That link was the whole of it. Through it, we assigned each patient a unique identification number. This unique ID did something very important. It tied the physical sample directly to the living person from the very start.

This connection remained solid throughout the entire lifecycle of the test. The ID bound the person to the sample from the moment of collection in the field. It stayed with the tube during processing inside our laboratory. It guided the data all the way through to the final returned result.

This was before the peak. We were doing wide screening, and the constraint was not the bench. It was the desk. Every patient's details were copied out by hand, and that copying set the ceiling for the whole laboratory. We could move fewer than three hundred samples a day, and no part of that limit had anything to do with virology.

Once the spreadsheet drove the barcode printer, each person carried a unique identifier and each tube carried it too. The identity was fixed at the desk, before anyone went to the field. That one change lifted us to roughly four thousand samples a day. I want to be exact about why, because the obvious reading is wrong. A label printer does not make a thermocycler faster.

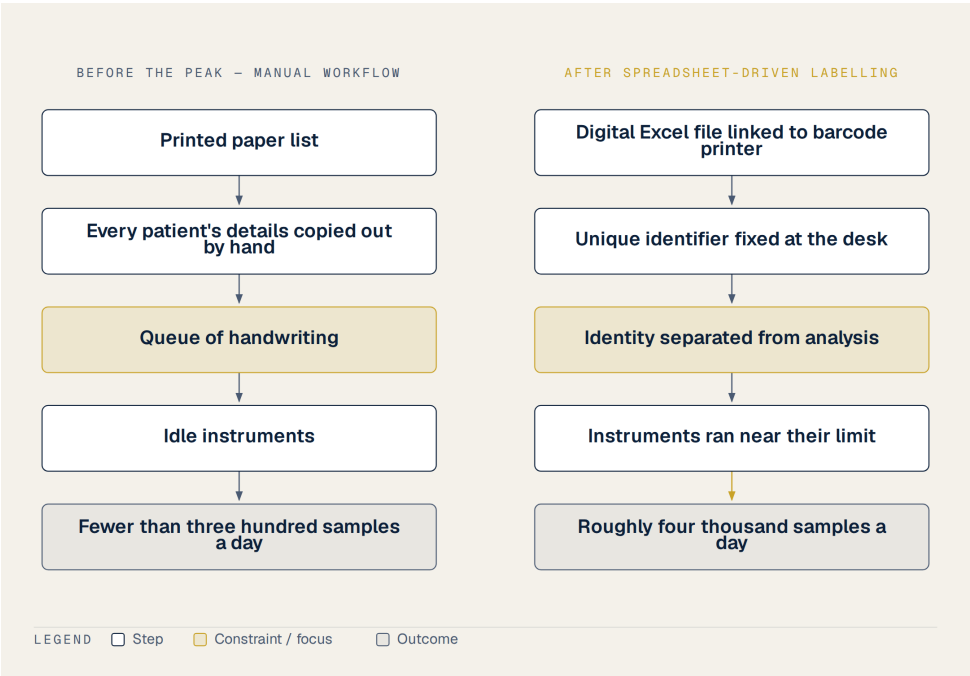
What it did was separate identity from analysis. With identifiers already assigned, we could pool samples into a single reaction and the record still held the list of who was in that pool, without anyone writing names on a worksheet. If a pool came back positive we broke it apart and ran the members individually, and nothing had to be typed in again. When positivity was low we pooled more. When it rose we pooled less, or ran singles. The testing strategy could change in a morning and the paperwork did not have to be rebuilt to follow it.



Identity is assigned at the desk, before anything is collected. Because the identifier was fixed before collection, the laboratory could change its testing strategy in a morning without re-building any paperwork.

At four thousand a day we were close to what our instruments could do. That is the honest ceiling, and it is the answer to the obvious objection. Removing the copying did not raise the analytical capacity of the laboratory by a factor of thirteen. It let us reach the ca-

capacity we already had. Before, the machines were idle behind a queue of handwriting. Afterwards they ran near their limit.



The bottleneck was the desk, not the bench. Analytical capacity was the same in both states. Removing the copying did not raise it. It let the laboratory reach it.

The identifiers did something else that mattered more than speed. We could attach whatever fields we wanted to a record and check them against each other, so a wrong birth year or a swapped tube had several chances to be caught rather than none. Speed was the visible gain. Fewer misidentified samples was the one that mattered.

I built it in a spreadsheet. Nobody procured anything. There was no vendor to wait for, so when the situation changed I changed it that same day, and there was no new budget line and no new staff. I want to be clear that this is not a story about my cleverness. It is a

story about how low the barrier already was in 2021, and it is lower now. The same thing today could be assembled far faster than I assembled it. The principle has not changed at all.

We did not hire any new staff. We had the exact same team of laboratory workers. We did not receive any new budget line to make this happen. The process was cheap, and it was entirely ours. It worked so well that the exact same process still runs at Vinh Long CDC today. It has survived because it fits the daily reality of a provincial laboratory.

When I look back on those hard days, I see a clear truth. I realized that our problem was never really about speed. We did not need to run faster. We did not need to work ourselves to exhaustion.

The real problem was the trace. The trace is the thread that connects a person to their sample. If you cannot guarantee that trace, speed is meaningless. It does not matter how many thousands of tests you run if you lose the connection between the person and the tube. The ID we created was not just a label. It was the physical anchor of that trace. It was the system that held our memory.

During those years, I watched my colleagues walk away. They did not leave because they hated the science. They left because they could not afford to stay. In the provincial public health system, the hardest people to retain are always the preventive medicine staff. I lived this reality every single day, and the numbers in the literature match exactly what I felt on the laboratory floor.

To understand why we lose these people, you have to look at how medical professionals make a living in Vietnam. A clinical doctor or nurse working in a hospital has an escape hatch. When their official shift ends, they can go to work at a private clinic. They can open their own practice or consult for private facilities. They have a source of private practice side income. But we do not have that op-

tion. A preventive medicine specialist cannot open a private clinic in the evening to track epidemics or test water quality. There is no private market for tracking diseases. Dieleman and colleagues documented this reality back in 2003. They found that the low salary and allowances at the commune and district levels were the main factors that destroyed motivation. Specifically, they noted that preventive medicine staff are the hardest to keep precisely because they lack the private practice income that clinical staff rely on (1).

The financial strain is not just a complaint. It is a documented barrier. In 2013, Tran and his co-authors studied health worker satisfaction at the commune level in Vietnam. Their findings were stark. Only 24.0 percent of the workers they surveyed were satisfied with their salary and allowances, which is to say nearly three quarters were not. The dissatisfaction went deeper than just the monthly envelope. Only 25.1 percent of those workers felt satisfied with their overall benefits. When it came to the physical tools of the trade, only 35.7 percent were satisfied with the available equipment. Even the working environment itself only managed a 41.8 percent satisfaction rate (2).

I remember sitting in our administrative room, looking at my colleagues, and knowing these percentages were real. We worked with outdated machines and struggled to find basic supplies. Our base pay was low, and there were no evening consultations to make up the difference. When you live on those margins, every month is a quiet struggle. This is why we are writing this proposal. We cannot simply expect people to stay out of pure devotion. The data has shown for decades that the lack of private practice income and the dismal satisfaction with salary, benefits, and equipment are draining our provincial teams of their best talent. If we want to secure our health system, we must address the invisibility of this work and

build a career path that keeps these specialists from walking out the door.

What we lost was not equipment. It was people. After the extreme strain of 2021, we desperately hoped for a period of stability. Instead, we experienced a quiet, massive departure. What happened next was a crisis of retention that changed the landscape of public health across our entire country.

The Ministry of Health counted 9,680 health staff who resigned or walked out between the first of January 2021 and the thirtieth of June 2022. The breakdown runs: 3,094 doctors, 2,874 nurses, 551 laboratory technicians, 276 midwives, 593 pharmacists, and 2,280 other public employees. Eight thousand eight hundred and ten of them worked under provincial health departments rather than for the ministry directly, which is to say they were the people furthest from Hanoi. The ministry listed four causes. The second was pay, and it named where the pay is worst: salaries and allowances in the public system are low, it said, and lowest of all at preventive medicine and grassroots facilities. That is the ministry describing the problem this book is about, in its own words, in 2022. These departures were not evenly distributed throughout our provinces. Instead, they were heavily concentrated in our largest urban areas where the pressure had been highest. Ho Chi Minh City alone lost 2,035 health workers during this period. Hanoi, our capital, lost 1,032. Even in our provincial laboratories, the vacuum was completely real. When colleagues left, their empty desks meant more night shifts and more exhausting work for those of us who stayed behind (3–5).

The government recognized this severe crisis and tried to respond. They chose the most direct tool they had available. In early 2023, they issued Decree 05/2023. This decree raised the occupational allowance to 100 percent for preventive health and grass-

roots medical staff. For a brief moment, it felt like our struggle was finally recognized. Our monthly income went up, and we hoped this financial support would help stabilize our depleted workforce (6).

However, this intervention had a significant catch that destroyed its long-term impact. The 100 percent allowance was not a permanent wage reform. The decree only applied this increased rate temporarily for the period from January 1, 2022, to December 31, 2023. Starting on January 1, 2024, the occupational allowance reverted back to the old, much lower levels set by the earlier Decree 56/2011. Money was tried here as a solution, but it was tried temporarily, and it did not hold people (6).

This temporary measure proved exactly what many of us knew from our daily lives. Financial incentives alone are not enough to retain staff when they are applied in isolation. It was a brief bandage on a deep, structural problem. It did not create a predictable career pathway, nor did it make our daily field labor visible. When the temporary allowance expired in 2024, the extra financial support vanished, and the old sense of professional invisibility returned. My colleagues did not just want a short-term cash injection to endure burnout. They wanted a sustainable professional future where their verified work is recognized. This is why we need a new approach, because temporary money has already failed to keep our specialists from leaving (6).

Many colleagues pack their bags and leave. When people talk about solving this retention crisis, they almost always point to money. They demand larger bonuses, higher allowances, or direct pay-for-performance schemes. As someone who lived on a low public salary, I understand this demand. Money is not wrong. For underpaid staff in developing countries, salary is not merely a hygiene factor that we can ignore. It is a matter of basic survival. However,

if we look closely at the global evidence on health worker retention, we find an honest and far more complicated picture.

The scientific evidence on using financial incentives to retain health staff is surprisingly thin and inconsistent. Consider the 2021 Cochrane systematic review by Diaconu and colleagues. They analyzed pay-for-performance schemes in low and middle-income countries. They found that these financial programs produced inconsistent effects, and they graded the certainty of this evidence as low to very low. Even the World Health Organization is remarkably candid about the limitations of its own guidelines. In its official recommendations on retaining health workers in remote and rural areas, the WHO states that almost all of its retention recommendations rest on low or very low certainty evidence. This low grading is due to research design limitations, rather than proof that the interventions do not work. For instance, their recommendation on providing social recognition for rural health workers is graded as strong, yet the evidence backing it is rated as very low certainty (7–9).

We must look at how health workers make choices when they are asked directly. Discrete choice experiments allow us to see what staff value. In Timor-Leste, a study by Rockers and colleagues in 2016 surveyed over four hundred workers. They found that professional skills upgrading and specialization actually ranked above wages as the number one preference. In Lao PDR, a 2013 study by Miyazaki and colleagues found a slightly different mix. There, salary alongside secure tenure and promotion dominated the preferences of health workers. These studies suggest that career pathways often matter as much as, or even more than, direct cash. But we must attach a vital caveat to these findings. These are discrete choice experiments. They measure stated preferences on a survey,

which does not always predict actual human behavior in real life (10,11).

We do not have randomized controlled trials comparing scholarships directly to cash rewards for frontline staff. What we do have is a clear pattern. Financial incentives alone are the weakest tool when applied in isolation. That is not an argument that pay does not matter. The Tran survey and the ministry's own attrition report both say plainly that it does, and a technician who cannot cover rent will leave no matter what certificate we hand them. A living wage is a precondition here, not an alternative. What the evidence suggests is that money paid once, during an emergency, buys presence for the length of the payment and nothing after it. The question this book takes up is what to build alongside the wage so that the years a technician spends at the bench leave something behind that belongs to them (2,5).

The foundation of global policy rests on the World Health Organization's 2010 global policy recommendations on retention in remote and rural areas and the 2021 guideline that replaced them. Both of these landmark documents are organized around four core pillars: education, regulation, financial incentives, and personal and professional support. Both guidelines explicitly state that coordinated, bundled packages of interventions beat single, isolated ones, and both conclude that relying on financial incentives alone is the weakest standalone bet available. The certainty caveat noted above applies to this conclusion as much as to any other in the set (8,12).

The Cochrane review conducted by Grobler and colleagues in 2015, registered as CD005314.pub3, evaluated the global literature and reached the exact same conclusion, finding very low certainty evidence for every single intervention type. We simply do not have high-quality trials to tell us what works with statistical confidence.

When we look for the most cited paper in this entire field, we find the 2008 systematic review by Willis-Shattuck and colleagues published in BMC Health Services Research, volume 8, article 247. Their classic analysis also found that financial incentives alone are insufficient to keep health workers at their posts. Instead, they identified recognition, career development, and management and supervision quality as the powerful, recurring themes that drive retention. However, we must remain completely honest about the nature of this classic study. It is a descriptive synthesis of the existing literature with no pooled effect sizes, meaning we cannot extract a single, unified mathematical impact from their work (7,13,14).

In our provincial laboratories, we have spent years acting as if salary bonuses are the only tool we have. But the evidence shows that professional recognition and a clear career pathway are the real missing links. By building a verified record of service, we are not trying to replace salary with symbolic gestures. Rather, we are addressing the exact non-financial motivators that the global literature has highlighted for decades. If we want our technicians to stay, we must make their invisible labor visible.

During the heights of 2021, I saw firsthand what a massive surge of international and national resources could achieve. For a brief period, laboratories across our country that had never coordinated before suddenly found themselves speaking a common, unified language. We were running comparable assays, reporting our diagnostic findings on comparable timelines, and connected to epidemiological networks that had not existed even a year before. This systemic interoperability was very real. It was incredibly expensive to establish, and we built all of it in a frantic hurry.

But nothing about this newly constructed infrastructure was permanent. An institution is an open system, not a sealed vault. A laboratory capability persists only while budget, staffing, public atten-

tion, and routine drills keep actively flowing into it. When that flow stops, the capability degrades. This is not a law of physics, and I do not want to dress it up as one. It is an ordinary administrative fact.

Our field already has a well-established name for this frustrating, repeating pattern. We call it the cycle of panic and neglect. The Global Preparedness Monitoring Board described this exact loop in 2019 as a process of ramping up national efforts when a threat is serious, only to quickly forget them once the danger subsides. Two years earlier, in 2017, the World Bank argued that pandemic preparedness must become a permanent, standing budget line rather than a series of emergency appropriations (3,4,15).

This neglect phase is not a distant forecast. It is measurable, and it is running right now. Global development assistance for health has fallen from roughly eighty billion dollars at its 2021 peak to about thirty-nine billion dollars in 2025. This is the lowest level of international funding in fifteen years, and it is forecast to fall even further. Meanwhile, the Pandemic Fund has awarded only about one and a half billion dollars in grants across three years. This stands against an estimated annual need of ten and a half billion dollars. Even the international pandemic agreement adopted in 2025 is still not in force because its technical annex missed its critical deadline (4,16).

We see the decay of these systems every day in our local units. The physical machines are still on our benches, but the software updates have stopped, and the trained technicians are leaving for other jobs. Notice which parts survived and which did not. The spreadsheet workflow cost nothing and belonged to us, and it is still running. The donor-funded sequencing network cost a great deal and belonged to a grant cycle, and it is going quiet. Cheap and locally owned outlives expensive and externally owned. That is the pattern

this proposal tries to sit inside, not an argument that a record can replace a budget.

I want to be exact about what attaching a name to a serial number does and does not do, because the claim is easy to overstate. It buys no reagents. It funds no service contract, patches no software, and pays nobody's salary. When the money stops, the machine still stops. What the record changes is narrower: it means that when a technician leaves, the demonstrated competence leaves with a document rather than with nothing, and that a laboratory rebuilding after a funding gap can find people who have provably done the work before. That is a retention and re-entry argument, not a financing one. Financing is somebody else's chapter.

The work we did vanished into aggregate administrative files. When we look closely, we find that the cycle of panic and neglect is not just a distant global pattern. It is a measurable, running reality in our own national backyard. This same problem of human invisibility repeats itself at a national scale in the Vietnamese preventive medicine budget.

In 2008, the National Assembly resolved that at least thirty percent of the state health budget must be dedicated to preventive medicine. This was a legally binding commitment, and later party resolutions repeated it. Fourteen years on, the delegation charged with checking that commitment reported what the money had actually done. Across forty one provinces and cities, the preventive share was twenty point four percent in 2012. In 2017 it was twenty two point four nine percent. In 2022 it was twenty four point four one percent. The direction is upward. The target was never reached. At roughly four tenths of a percentage point a year, the gap left in 2022 would take another fourteen years to close, which would put the country at a target it set for itself in 2008 somewhere around 2036. That last figure is my own arithmetic on three published

points, not a government projection, and the reported figures are drawn from spending on grassroots health in forty one provinces, out of the sixty three that existed when the review was done, so they are not strictly the same denominator the resolution names. (17–19)

Nobody broke a law here. There is no villain in this story, no scandal, no diverted money. There is a promise a country made to itself, in writing, and a system moving toward it so slowly that a technician who started work the year the resolution passed would retire before it arrived.

The more revealing problem is that the delegation could not settle on the number at all. Its own chair raised two objections in the same breath. She questioned what should even count as preventive spending, and she noted that the provinces report the ratio differently from one another. She asked the Ministry of Health for a more specific report so that the delegation would have something solid enough to write into its findings. The body with the constitutional authority to audit the government could not establish, to its own satisfaction, how much the government spends on preventing epidemics. Not because anyone hid it. Because the work was never recorded in a form that survives the trip from a district health centre to a national ledger. (19,20)

Then the ledger itself moved. On the twelfth of June 2025, Vietnam reduced its provincial administrations from sixty three to thirty four: twenty eight provinces and six cities. Nineteen provinces and four cities were formed by merger; eleven were left alone. This was the largest administrative consolidation in decades, and it was carried out for sound reasons that have nothing to do with this book. But consider what it means for a record. Two provincial health systems become one. Two sets of registers, two numbering conventions, two server rooms, two ways of deciding what counts as preventive spending, are pushed together into a single new au-

thority with a new name. Every institutional archive in the country was rewritten in the same season. (21)

Institutions survive that kind of change. They have lawyers and letterheads and successor authorities. A technician does not. When the office that employed her for nine years ceases to exist, the evidence of what she did lives wherever her old supervisor filed it, in a building that now answers to a different province. She cannot subpoena her own past. This is the sharpest argument I can make for a record that belongs to the worker and travels with her. Reorganisations are not rare accidents. They are a normal feature of a functioning state, and they will happen again. A career should not have to survive them by luck.

The honest thing to write here is that the number is simply not published. I will not guess at it or estimate it, because the absence of the data is itself the more useful finding. A national system that cannot say how much it spent on prevention is a system in which prevention work is not recorded in a form that anyone can check. These are not the same problem. Fiscal classification is a matter of accounting standards; a technician's missing logbook is not. But they rhyme, and they rhyme for the same reason: in both cases the work was done and the record of it was never built to travel. In our laboratory, a technician's labor was real, but because it was never recorded in their name, it remained invisible. At the national level, the failure of our accounting system produces the exact same result. The preventive work of our clinics and laboratories is real, but the record of it is not. If we cannot track the money we promised to spend, we cannot hope to track the pathogens we are trying to stop.

Timekeeping or Scorekeeping

The administrative office of our provincial CDC is quiet at five in the morning. On the desk, a printed list of names lies next to a keyboard. A stack of blank transport medium tubes sits in their plastic racks, clean and empty. Beside them, the barcode printer waits, its green status light blinking slow. I am wearing my ordinary clothes, a faded cotton shirt and trousers. There are no clinical samples in this room yet, so there is no need for a hazmat suit, heavy blue gown, or double layers of gloves. But as I look at the paper list under the fluorescent tube, a quiet realization comes over me. This desk work is the most dangerous step of the entire pandemic response.

I open the email that arrived an hour ago. The messaging app on my phone is already buzzing with new lists sent by district teams. There are names of citizens quarantined in facilities after arriving from overseas, names of prisoners from the provincial jail, and names of people who died overnight in their homes. I begin transcribing them. My fingers move across the keys, converting the text into a structured Excel file. I connect the file to the barcode software and hit print. The machine hums, spitting out small, sticky labels with black-and-white lines. I peel each one and wrap it around a blank plastic tube. This barcode is a unique identifier, a digital thread that will bind this patient's identity to their specimen from the moment of collection to the final diagnostic result.

Outside this office, everyone associates our response with the drama of biological hazards. They think of the ambulance engine

starting up, or whatever vehicle our unit could find to transport the teams. They think of the cold boxes, foam or ordinary ice, waiting in rows along the wide corridor outside. Yet, we cannot go out to collect a single sample until these labels are ready. If a barcode is printed with a wrong digit or stuck to the wrong tube, a patient's result is lost, and the chain of containment breaks.

We used to move fewer than three hundred samples a day, and the ceiling was this desk, not the bench. With identifiers fixed here before anything goes to the field, the same team can put four thousand through the day. The arithmetic is not mysterious and it is worth setting out, because a barcode printer plainly does not make a thermocycler faster. Each reaction well holds one tube, so when every person needs their own well, the number of people tested in a day can never exceed the number of wells run. Pooling breaks that equivalence: several people share one well, and one run now covers several times as many people. What had made pooling impractical was the paperwork, since somebody had to write out by hand which names sat in which pool and then write them out again to retest a positive one. Once the identifiers were fixed at the desk, that writing disappeared, and the pool ratio became a dial we could turn against the positivity rate that week. The throughput did not come from the printer. It came from removing the clerical work that had made the instruments' real capacity unreachable. The ceiling moved off the desk and onto the bench, where it belongs. Same staff. No new budget line. This administrative routine is still running in our department today. I finish pasting the last label. The plastic tubes are now ready to be packed into the cold boxes. I look down at my hands, still free of gloves. The real bottleneck of the crisis is not happening in a secure laboratory or a quarantine ward. It is happening here, at a wooden desk, in everyday clothes, under the hum of an office light.

In those five years, I did not just handle samples. I watched how people responded to the systems around them. During my time in the laboratory, I observed a fundamental truth about data. The exact same set of records can produce two completely opposite human reactions. It all depends on who owns that data. Under one frame, which we can call the timekeeping mindset, workers feel watched and monitored. They see the system as an intrusive eye. It is a tool of surveillance that tracks their minutes and counts their mistakes. When people feel controlled, they do not want to cooperate. They hide their efforts, they resist, and they do only the bare minimum required to avoid trouble. But there is another frame. If you shift the design so that workers are building their own records, the entire dynamic changes. Under this framework, the data belongs to them. It is their point-accumulation system, a physical ledger of their own sweat and labor. When workers know that they own the record of their own work, they do not hide. They actively step forward to volunteer the evidence of their achievements. They want to log every hour, every field visit, and every test because they are building a professional asset that belongs to them.

This dramatic shift in behavior is not just an observation from our laboratory floor. It is explained by Self-Determination Theory, a psychological framework that highlights our basic needs for autonomy and competence. While researchers consider the core psychological mechanism of this theory to be strong, they rate its direct application in workplace systems as moderate. According to this theory, the way we frame a system changes everything. The theory draws a sharp distinction between informational rewards and controlling rewards. Informational rewards are those that confirm our competence and acknowledge our skills. When a worker receives feedback that confirms their mastery, their intrinsic motivation rises. They feel a sense of pride and professional growth. Controlling

rewards, on the other hand, are designed to pressure, monitor, and push people. They are tools of control that signal to the worker that they are under constant surveillance. When a system relies on controlling rewards and pressure, motivation drops. This psychological reality tells us something vital about system design. The way we frame a system is not a minor cosmetic detail. It is a core design variable. How we structure the flow of data is what determines whether the system succeeds or fails.

In my proposal, I am not asking for a more expensive system to watch our staff. I am proposing that we change the frame entirely. We need to stop using data to monitor people and start letting people use data to build themselves up. When we make this shift, we transform invisible field labor into a verified professional asset. This is the difference between a system of surveillance and a ladder of opportunity.

I am writing this proposal because I know what it is like to work in the shadows, where your sweat has no physical record. Many people think that motivating health workers requires complex schemes or expensive campaigns. But as I sat at my desk and looked at the global literature on human behavior, I discovered the strongest evidence for our proposal. It shows that the simple act of tracking your own progress is a powerful force.

This is not a vague claim. In 2016, Harkin and his colleagues published a landmark meta-analysis in *Psychological Bulletin*. They compiled data from 138 independent studies, representing roughly 19,951 participants. Their findings were clear. They found that monitoring progress alone raised goal attainment. It did so with a substantial effect size of 0.40. But there was a vital detail in their findings that speaks directly to my experience in the laboratory. They discovered that this positive effect was even larger when progress was physically recorded. It was also larger when that

recorded progress was reported or made public. This is the true pillar of our entire design. The value of this system does not lie in trivial gamified badges or superficial point systems. The value lies in the physical, verified recording of a worker's daily labor. It is the act of turning invisible actions into a permanent record (22–24).

One limit belongs with that finding before I lean on it. Harkin and colleagues were synthesising studies of people pursuing their own goals: losing weight, giving up cigarettes, keeping to a course of study. Nobody in those trials was building a record that an employer, an auditor or a scholarship board would later read. Carrying the result across to a high-stakes workplace log is an extrapolation. The mechanism is plausible and the direction is probably right, but it has not been tested where the record decides something about the person keeping it. (24)

We must look at how this physical record changes how a worker feels about their employer. In our field, people walk away because they feel completely invisible and unsupported. This feeling is captured by a concept called Perceived Organizational Support, or POS. In 2017, Kurtessis and his colleagues published a massive meta-analysis in the *Journal of Management*. They looked at how this sense of support correlates with employee behavior. The numbers they found are remarkably strong. A worker's perceived organizational support has a massive positive correlation of 0.69 with their commitment to the organization. When staff feel supported, their intention to leave drops significantly, with a negative correlation of minus 0.50. Their actual rate of quitting drops as well, with a correlation of minus 0.21 (25).

But how do we build that sense of support in a provincial laboratory? The meta-analysis by Kurtessis and his team found that development opportunities are among the strongest sources of perceived organizational support. Giving people a chance to grow is far more

powerful than temporary bonuses (25). This is why our proposal links a verified record of service to tangible educational gateways like scholarships. We are not just printing labels or tracking hours for surveillance. We are creating a physical record of competence that a worker can use to claim a professional future.

That sentence contains the most dangerous assumption in this book, and I would rather name it than let a reviewer find it. A ledger does not create a scholarship. It records what somebody did; it does not fund a place for them to study. If a laboratory installs the record, tells its technicians that verified service opens a door, and then no door opens because no institution ever agreed to honor it, we will have built the precise broken contract that Rousseau describes, and we will have done it deliberately, with better record-keeping than before (26). That outcome is worse than doing nothing, and it is the likely outcome if the sequence is run in the wrong order.

So the sequence matters more than the software. The scholarship commitment has to exist first, in writing, from a body that can actually admit students, before a single technician is told that the log leads anywhere. Until that letter exists, the record should be described to staff as exactly what it is and nothing more: a document of their own work, useful for their own curriculum vitae, owned by them. That framing is honest at any funding level and survives the pilot failing. Anything beyond it is a promise, and a promise is not ours to make until somebody with a seat at a training institution has signed for it (26). When we do this, we transform the daily grind into a visible ladder of growth, backed by the strongest psychological science we have (25).

At the time I often heard suggestions from visiting consultants about how to motivate our staff. They frequently proposed setting up leaderboards. They wanted us to display a public ranking of who

processed the most sample tubes or who logged the most hours in the field. They believed that public competition would naturally drive us to work harder and faster. In my proposal, however, you will find no leaderboards. I have intentionally excluded them from the design of this recognition pathway. This decision is not a matter of personal taste. It is guided by what actually happens when you introduce ranking systems to frontline health workers.

We should look at the real-world evidence from field research. A recent study on the design of gamified training systems for community health workers found something very clear. Frontline workers do not read public ranking systems as a source of healthy motivation. Instead, they read them as tools of surveillance. When you introduce public scores, it creates a high level of anxiety. It breeds resistance and tension inside a professional environment where people must rely on one another to survive. In our laboratories and field units, we do not work in isolation. We work in tight, interdependent teams.

Furthermore, the broader literature on gamification supports this caution. A systematic review by Sailer and Homner in 2020 analyzed the actual effectiveness of these game-like elements. They found that competition on its own produced smaller gains in learning outcomes and in behavioural measures than competition paired with a cooperative element, and I should be precise that those are the outcomes measured, not retention or accuracy at a workbench. When you force people to compete against their colleagues without any cooperative element, the team dynamics break down. Even more importantly, their meta-analysis showed that when you look at high-rigor studies, most of the grand claims about gamification begin to fade. They looked at cognitive, motivational, and behavioral effects. Only the cognitive effect of gamification actually held up under high-rigour analysis. The behavioral and emotional

changes that marketers promise simply do not survive strict scientific scrutiny (27).

We must be honest about what we do not know. There is currently no randomized controlled trial evaluating the use of points, badges, or leaderboards for community health worker performance in low and middle-income countries. The trials simply do not exist. What has been documented to work in these settings is financial incentives tied directly to verified activities. Public rankings, on the other hand, are what consistently backfire.

For these reasons, the pathway I am proposing relies on self-competition. People should compete only against their own past record, not against their colleagues. A worker's dashboard will show their own progress, their own verified hours, and their own path toward their educational goals. It will not show where they stand in a public hierarchy. We want to build a tool that confirms competence and records sweat, not a scoreboard that publicizes shame. When we remove the leaderboard, we remove the feeling of being watched, and we allow workers to focus on their own growth.

I filled out my share of paperwork. We would often sit down at the end of the month, try to remember what we had done weeks earlier, and write it down on paper sheets. But as I write this proposal, I must be completely honest about this practice. A paper log submitted at the end of the month is not actual scientific evidence of work. It is a statement. It is a retrospective claim, a recollection of events that is highly prone to human error and fading memory. Real data integrity requires something much more rigorous.

The global pharmaceutical and medical industries have already named the standard we must follow. It is called ALCOA. This standard was originally coined by Stan Woollen at the United States Food and Drug Administration in the early 1990s. He created it as a simple mnemonic device to train inspectors. Over time, through

widespread adoption across clinical trials and manufacturing, this mnemonic became the global gold standard for data integrity (28).

It is important to note a vital caveat here. These standards were written for regulated pharmaceutical manufacturing and clinical laboratories. None of them were originally designed for a provincial preventive medicine lab in Vietnam. Applying them in our provincial CDC is a deliberate borrowing of an industry standard, not a legal requirement. Yet, if we want our records to be trusted internationally, this is the bar we must meet.

The very first letter in this standard stands for Attributable. To be attributable, a record must show exactly who performed an action and when they did it. This information must be tied directly to a unique individual identity. In many provincial laboratories, staff often share a single computer login or sign off on sheets using a generic team name. This is a classic inspection finding because shared logins completely break accountability. If everyone is logged in as the same user, the trace is gone. You cannot verify who actually ran a test or who made a change.

The third letter in the standard stands for Contemporaneous. This means recording the data at the exact moment the work is performed. Writing logs late, keeping rough notes on scraps of paper to copy them later, or batching data entry at the end of the day is not a minor paperwork error. In the eyes of international inspectors, it is a direct violation of data integrity.

This is why we need to build a digital pathway that records work as it happens. When a technician scans a barcode on a container tube, the database must log that action instantly with their unique identity. We cannot rely on retrospective paper statements compiled weeks after the fact. I have to secure a contemporaneous, attributable record. That is the only way to transform our invisible

daily sweat into a verified professional asset that outside institutions can trust.

I saw how easily human memory fails when it is forced to do paperwork weeks after the fact. In this proposal, I want to address the third core letter of the ALCOA standard: Contemporaneous. In the world of high-standard data integrity, contemporaneous means that an activity must be recorded at the exact moment it happens. It cannot wait. When recording happens hours or weeks after the work, certain practices appear everywhere, and they appear for the same reason. Logs get completed after the fact. Numbers get written on whatever paper is at hand and transcribed into the official ledger later. Entry gets batched to the end of a long day. None of this comes from dishonesty. It comes from a system that was designed to make people remember instead of designed to let them record. Under international data standards these practices are treated as data integrity failures, and the standards are right about the consequence: the record stops being trustworthy. But the fix does not belong to the person holding the pen. It belongs to whoever designed the moment at which the pen is picked up.

To prevent these failures, we must design a system that complies with the highest regulatory rules, like the United States Food and Drug Administration's 21 CFR Part 11. Specifically, section 11.10(e) requires a computer-generated, time-stamped audit trail that records every creation, modification, or deletion. The rule states that any change must not obscure previously recorded information. Compliant laboratory systems do meet that rule, with append-only audit tables, database controls and digital signatures. What they generally do not add is cryptographic immutability that holds against the system's own administrator, and no rule requires them to. The audit trail answers to the institution that runs it. This means that if we build a system where a record is pushed immedi-

ately to a server and secured unalterably using cryptography, we are actually performing above the current regulatory standard rather than merely catching up to it.

There is a limit to what this technology can do, and it is worth naming before anyone gets excited about cryptography. A tamper-proof log of fabricated entries is still a tamper-proof lie. Cryptography cannot prevent a technician from lying about a test they never ran.

Because of this limitation, the whole idea of our digital design is mistake-proofing. In manufacturing, they call this poka-yoke. We should not waste our time telling stressed, overworked frontline staff not to do things wrong. We should not rely on constant warnings or heavy training manuals. Instead, we should build a workflow where the easy path is the correct one. When a technician scans a barcode, the time and the identity are fixed at that moment, so writing the entry up from memory a week later stops being the convenient option. I want to be exact about the limit here, because it matters. A scan proves that a label was read at a particular second. It does not prove that a specimen was drawn, and nothing stops somebody scanning fifty tubes at a desk. Contemporaneous capture removes the drift that comes from writing things down later. It is not a defence against someone who has decided to lie, and the later layers exist because of that. By securing the data at the moment of creation, we make integrity the only path forward.

I also learned that you cannot build a sustainable health system on guesswork. When we design tools to motivate frontline workers, we must rely on honest, proven mechanics from behavioral research rather than marketing hype. Our proposal does not try to invent human behavior. Instead, we want to look at how people actually respond to progress and thresholds, using established science while remaining completely honest about its limits.

The first behavioral concept we must build upon is the endowed progress effect. In 2006, researchers Nunes and Drèze published a field experiment studying how people complete tasks. They compared two scenarios. In the first, people received a loyalty card with eight blank spaces. In the second, they received a ten-space card that already had two pre-stamped marks. Technically, both groups needed to complete exactly eight tasks to reach the goal. Yet, the group with the pre-stamped cards showed a much higher completion rate, rising from 19 percent to 34 percent. The feeling of already being on a journey is a powerful psychological driver. However, we must preserve a vital caveat from this study. This pre-stamped progress only works if there is a stated, logical justification for the endowment. If you give people points arbitrarily without explaining why they earned them, the effect weakens significantly. Furthermore, there has been no strong, independent replication of this experiment. We should trust the general direction of this effect, but we must never treat these exact percentages as absolute laws (29,30).

The second mechanic is the goal gradient effect, documented in 2006 by Kivetz, Urminsky, and Zheng. Using multiple research methods, they proved that people naturally accelerate their efforts as they get closer to a threshold. In their study, participants accelerated their rate of progress by roughly 20 percent as they neared the goal. But their research also documented a critical post-reward slump. Right after a person receives their reward, their motivation drops, and their progress slows down significantly. This post-reward slowdown is a major risk for health systems. If we build a system with a single, one-shot threshold, our staff might work hard to reach it and then immediately stop. To prevent this, we must design continuous thresholds rather than one-shot targets. The ladder of

professional development must keep going so that there is always another step within reach (31).

Finally, we must acknowledge the ethical boundary of these mechanics. These effects were originally designed to sell consumer goods and increase retail purchases. The right way to apply them in public health is to help workers see their own hard work accumulate as a genuine ladder of growth. They must never become a device for extracting unpaid labor from underpaid staff. By combining these honest mechanics, we can create a system where workers feel they have already started their journey, and where they can safely build a professional future.

I am writing this proposal because I know how a poorly designed system can crush a team. When people talk about gamifying public health, they often suggest using daily streaks. They want us to log in every single day to keep a digital streak alive, just like popular language learning apps do. But in our proposal, we have completely rejected hard daily streaks. This is a deliberate choice based on both the realities of provincial field work and an honest look at behavioral evidence.

A daily streak system is actively punishing for the people who actually do the work. In a provincial CDC, our schedules are never predictable. We do not work a neat, daily nine-to-five routine. We work exhausting night shifts, and we get pulled into sudden, urgent outbreak responses that last for days. When you are standing in a remote village trying to contain an epidemic, you do not have the time or energy to log into an app just to save a digital streak. Missing a single day because of a real emergency ruins your streak. It wipes out weeks of visible progress. Instead of motivating people, it makes them feel defeated. Because of this, our design throws out daily requirements. We use weekly quotas and forgiveness tokens

instead. This gives staff the flexibility to manage their intense, unpredictable workloads without feeling penalized by the system.

The evidence behind daily streaks is thinner still. If you look at the widely quoted statistics about how people lose motivation or quit when a streak is broken, you will find a surprising truth. Those numbers trace back only to marketing blogs. They have no verifiable, peer-reviewed source. It is mostly industry lore, not established science. We cannot build a state system on marketing myths.

This brings me to the second decision. We have structured the data around ALCOA, and I said earlier that borrowing it is a choice rather than an obligation. It is worth saying why we would choose a standard nobody is holding us to. We choose to borrow it because it is the only way to build a record that international partners can trust, but we must acknowledge this choice openly. We are not catching up to a legal mandate. We are choosing a higher path because our staff deserve a record of their labor that is too rigorous for anyone to dismiss. By combining realistic weekly goals with strict data integrity standards, we can build a system that respects the lives of our workers while protecting the truth of their work.

Running assays under pressure taught me how easily ideological debates about motivation can lose touch with reality on the ground. When we design a system to recognize and reward our technicians, we must address the motivation debate honestly rather than picking over extreme academic theories. For decades, experts have argued about whether extrinsic rewards destroy a person's inner drive. In 1999, Deci, Koestner, and Ryan published a classic meta-analysis showing that tangible, performance-contingent rewards actually reduce free-choice intrinsic motivation, with effect sizes between -0.28 and -0.40, while verbal praise increases it. This study became a major argument against using physical rewards in the workplace. However, the debate did not end there. Gerhart and

Fang later counter-argued in 2015 that the actual field evidence for this supposed harm is remarkably thin. They pointed out that laboratory experiments do not reflect the complex, long-term realities of a functioning public health organization (32,33).

The best current reconciliation of this conflict comes from a massive meta-analysis by Cerasoli and colleagues in 2014, published in *Psychological Bulletin*. Drawing on forty years of work and 183 separate samples, their work shows that intrinsic motivation and external incentives are largely additive rather than opposed. They do not cancel each other out. Instead, they drive different aspects of performance. Intrinsic motivation is a powerful predictor of work quality, while incentives are a strong predictor of work quantity (34).

Therefore, any defensible claim we make in this proposal must remain narrow. We have to reject the dramatic overclaim that money simply destroys professional motivation. That statement is flatly incorrect and unsupported by modern evidence. What the evidence actually shows is that output-contingent money narrows human effort toward what is actively measured, and this focused effort is incredibly fragile once those incentives are withdrawn. In a provincial laboratory, if we only pay for the speed or number of tests processed, our technicians will focus entirely on those metrics and ignore vital duties like instrument calibration, biosafety cleaning, or detailed documentation. Once that temporary money stops, the motivation to perform those measured tasks collapses entirely.

The same absence of trial evidence noted above covers these motivation models too. We are borrowing frameworks from corporate and clinical research in wealthy countries to design tiered scholarships and verified service logs for a Vietnamese provincial laboratory, and the borrowing is the weak joint in the argument. We must not overpromise. By understanding that intrinsic care and structured incentives work together, we can design a registry that honors

a technician's dedication to quality while offering clear, tangible paths for their professional growth.

A well-intentioned gesture of recognition can easily turn toxic. We are proposing a digital registry to track and reward the field contributions of our technicians, but we must understand the precise design conditions that decide whether such a system builds or corrodes our workforce. This is not a matter of administrative preference. It is a matter of documented human behavior, and we must design with these behavioral realities in mind from the very first day.

The first critical condition comes from organizational justice research, specifically the work on procedural fairness by Colquitt and colleagues in two thousand and one. Their research shows that the fairness of the process used to decide who is selected for a reward determines its ultimate impact on the workforce. If the selection process is opaque, secret, or perceived as biased, the reward ceases to be an incentive. Instead, it turns directly into a grievance for those who were passed over, destroying organizational trust and eroding cooperation in the laboratory. For a provincial CDC where teamwork is a matter of biosecurity, an opaque reward is a dangerous hazard. (35)

The second condition is defined by psychological contract research, particularly the framework by Rousseau in 1995. The evidence here is clear and moderate in certainty: promising a professional pathway and then failing to deliver it produces psychological harm that is far worse than never making the promise at all. When an organization violates this psychological contract, staff members experience deep resentment and a collapse in motivation. If we build a registry that promises our technicians a gateway to scholarships, but those gateways are blocked by bureaucratic delays or in-

stitutional failures, we will actively destroy the fragile commitment of our frontline staff. (26)

Finally, we must ground our registry in basic needs research, which forms the core mechanism of Self-Determination Theory as outlined by Deci, Olafsen, and Ryan in 2017, and Van den Broeck and colleagues in 2016. While the theoretical mechanism is strong, its practical application carries a moderate level of certainty. This research demonstrates that three basic psychological needs, which are autonomy, competence, and relatedness, each independently predict employee well-being and job attitudes. Crucially, the organizational context itself predicts whether these three needs are met. This means the system around the record matters far more than the record itself. A digital ledger is a tool. If the system surrounding it is coercive, controlling, or competitive, the tool will fail to support our staff (36,37).

I must be completely honest about these findings. These behavioral theories are well-documented in international literature, but there is currently no trial evidence or randomized controlled study evaluating how they apply to preventive health workers in low-resource Vietnamese provincial settings. We are proposing a logical, evidence-informed design to test these hypotheses. By ensuring procedural fairness, honoring our psychological contracts, and focusing on basic psychological needs, we can build a supportive scaffold rather than a corroding trap.

In writing this proposal, I must present the real evidence regarding professional recognition and performance tracking, looking specifically at what works for health workers rather than consumers. Many digital platforms promise miracles based on studies of online shoppers or language learners, but our clinical reality is entirely different. When we examine the actual evidence from healthcare settings, the picture is incredibly narrow.

The strongest single trial evaluating gamified continuing medical education was conducted by Kerfoot and colleagues in 2014. This study ran for fifty-two weeks and tracked more than fourteen thousand patients. Even in this best-case scenario, the trial found only a statistically significant but very small improvement in the time it took to achieve blood pressure control. This is what the absolute ceiling of gamified professional education looks like under rigorous trial conditions. It is a modest, incremental gain, not a revolutionary transformation. (38)

Furthermore, when we look at the systematic review of thirty studies in health professions education conducted by Gentry and colleagues in 2019, the limitations of the current literature become even more obvious. Out of those thirty studies, only two evaluated gamification as such, rather than a single game-like element bolted onto a course. The rest were serious games or simulations. More importantly, the vast majority of the participants in these studies were medical students or residents rather than practicing clinicians who work under the daily, high-pressure constraints of a provincial clinic. We have almost no direct evidence showing how these digital tools affect the behavior of seasoned professionals. (27,36,39)

The gap is widest when we look at community health workers in low-resource settings. Kok and colleagues, reviewing what makes community health worker programmes work across low and middle-income countries, found that predictable packages combining financial and non-financial elements outperform either alone for both retention and performance (40). The review also indicated that publicly displayed recognition improved service delivery, which sits awkwardly beside the refusal of leaderboards set out above, so I should say where the line falls. Recognition that is public and absolute is welcome: a certificate, a named record, a colleague knowing what you did. Recognition that is public and comparative is not: a

ranked list that tells a technician they are seventh. The first says the work was seen. The second says somebody else was better, which is a different message and, in the studies cited above, a corrosive one. This is the very narrow, moderate band of evidence that our proposal actually stands on. We are not claiming that digital badges or leaderboards are a proven magic wand. In fact, we must be entirely honest. There is currently absolutely no trial evidence or randomized controlled studies evaluating the impact of leaderboards, digital points, or gamified badges on community health workers in low-income or middle-income countries. Indeed, some design studies suggest that public leaderboards can cause anxiety and destroy collaboration among staff.

Therefore, our proposal is not a grand, speculative bet. We are proposing a modest, highly contained twelve-month pilot of a verified service record to test these behavioral hypotheses in our local context. If we keep our expectations bounded and our design grounded in the real, limited evidence, we can build a functional tool that supports our technicians without overpromising. We need to build on what is proven, not what is hyped.

Three Readers, One Record

At three o'clock on a rainy Tuesday afternoon, three people are looking at the exact same line of data on their screens. In our provincial laboratory in Vietnam, a technician named Vy stares at a row of text showing forty-eight polymerase chain reaction cycles completed at midnight. For her, this line is her only shield, the physical proof of her ten-hour night shift. Down the hall, her supervisor looks at the same line because a query about a contaminated batch just arrived. For him, this row of text is a liability map, a way to find who is answerable for the error. Thousands of miles away in an air-conditioned office, a program officer opens a database. He has never seen our cramped room, but he paid for the reagents. For him, this line is a consumption metric, proof that the materials did not expire in a warehouse. All three are reading the same entry at the same moment. Yet, they see three entirely different realities.

Our data served the institution, the funder and the auditor. It did not serve the person who produced it, and nobody designed it that way on purpose. Modern laboratory informatics has built sophisticated tracking systems that view attribution purely as an instrument of blame and auditing. In Vietnam, thirty-eight laboratories have implemented open-source LIMS with US CDC support. However, these systems are managed provincially and lack individual, machine-enforced audit trails. When something goes wrong, we search the logs to find who to punish. When something goes right, the technician's labor remains completely invisible.

Our proposal is to turn this shared line of data into a voluntary point-accumulation asset owned by the worker. Rather than using it as a top-down surveillance tool, the technician can export her own verified contributions as a portable professional credential. This shift from surveillance to self-owned progress is supported by strong behavioral science. Meta-analyses on Perceived Organizational Support show that development opportunities are among the strongest drivers of employee commitment.

It is necessary to design this registry carefully to avoid the traps of gamification. For instance, studies on status tiers show that demoting an individual damages loyalty, so our system will never demote a user. We also reject daily streaks and point expiration, which behavioral literature shows can lead to data gaming or burnout. Instead, we rely on established concepts like endowed progress, where providing an initial, justified jumpstart on a progress record has been shown to increase completion rates from nineteen to thirty-four percent in field experiments.

However, we must remain completely honest about our limits. While behavioral mechanisms like endowed progress and goal gradients are well-documented in marketing studies, their application to healthcare is mostly untested. There is currently absolutely no trial evidence or randomized controlled studies evaluating how a worker-owned, point-accumulation registry affects reporting accuracy or staff retention in low-resource public health laboratories. We are proposing a logical, evidence-informed pilot of twelve months to test these ideas. By connecting the machine's serial number to a technician's name, we can satisfy the supervisor's need for accountability, the funder's need for tracking, and the worker's need for visible, portable credit.

Many digital systems and data initiatives come and go. Most of them arrived with a great deal of noise, only to end up abandoned.

They died because of a fundamental flaw in their architecture. They were built for only one party. In the world of public health, if you design a system that serves only a single group, you are writing its death sentence.

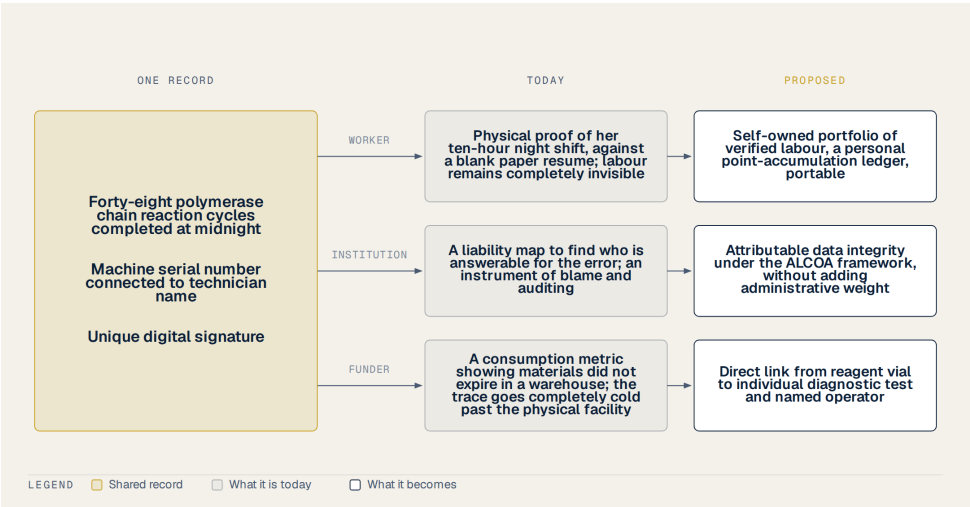
Most of the tools we were forced to use were built entirely for the funder. The international donors or the high-level administrators wanted clean reports and compliance metrics. Because the system was designed only for their eyes, we on the laboratory floor felt policed and monitored. We saw the software as an intrusive eye that tracked our minutes without offering us any value. When staff feel watched, they naturally resist. They find ways to work around the system, or worse, they begin to fabricate numbers just to make the charts look green. The data becomes a performance, and the system dies of its own falsehood.

Conversely, you can try to build a system solely for the worker. You can design an app that helps us track our daily tasks or manage our own professional learning. But if the system serves only the worker, it hits a different wall. Nobody wants to pay to run it. Government agencies and international donors will not fund a platform that does not provide them with verified, actionable data. Without their financial support, the technology cannot survive, and the project quickly collapses once the initial enthusiasm fades.

The proposal does not choose between those readers. It rests on one observation: the same record can serve all three of them at once.

For the worker, the record is a self-owned portfolio of verified labor. It is their personal point-accumulation ledger that they can carry with them to unlock professional opportunities, like scholarships. For the funder, this very same record is trusted, tamper-evident proof of activity. They can see that the tests were actually run, satisfying their need for accountability without forcing us to feel po-

liced. For the institution, the third reader, the same record answers the question of who did what on which day, which is the question its own regulator will eventually ask it. Further downstream sits a fourth party who is not one of the three but matters to all of them: the educational partner or licensing body, which can read a record it did not have to trust on faith. That reader is the reason the record has to be portable rather than merely tidy.



One record, three readers. The same row of data, read by three parties who currently want different things from it. Nothing new is collected; only who may read it changes.

We should be entirely honest about the limits of this design. A tamper-proof log of fabricated entries is still just a tamper-proof lie, and cryptography does nothing about what a person types in the first place. I made a claim in an earlier draft that I have since decided is wrong: that aligning the three readers around one record removes the incentive to cheat. It does the opposite. So long as the log was only an institutional liability record, a technician had little to gain from inventing runs. The moment those lines convert into

scholarship tiers and portable credentials, every entry becomes personally valuable, and the incentive to inflate counts, log runs that never happened, or work under a colleague's login rises accordingly. Making the record worth something necessarily makes it worth faking. Supervisor co-signature and spot-audit are the controls we have, and both are vulnerable in exactly the setting this proposal targets, where the supervisor countersigning may be the same person whose unit is measured by the total. Any honest pilot has to treat verification fraud as a live failure mode to be measured, not a risk that good design has already dissolved. We transform a tool of surveillance into a mutual asset.

Our staff work under a system that makes their daily labor invisible. When we write logs or enter data, we are creating records that could serve multiple purposes. Yet, most systems fail because they are built to be read by only one party. In this proposal, I want to describe a system designed for three readers at once, using a single digital record.

The first reader of this record is the worker. Today, if a technician spends five years running thousands of PCR tests in a provincial lab, they leave with nothing but a blank paper resume. They have no portable proof of their work. I am proposing a self-owned digital portfolio that is a permanent, verified record of their actual labour. It is a record that says, for example, I performed four thousand two hundred verified diagnostic tests over these specific years under a unique digital signature. This is not a symbolic badge or a superficial points system. It is a portable professional asset. It belongs entirely to the worker, and they can carry it with them when they move between different employers or apply for advanced training.

The second reader of this record is the institution. In our current provincial labs, there is often deep confusion about who did what

on any given day. To solve this, our proposal borrows the strict data integrity standards of the international pharmaceutical industry, specifically the ALCOA framework. In these standards, the very first letter stands for Attributable. This means a record must show exactly who performed a test and when they did it, mapped against a unique digital identity. When an institution reads our proposed database, they gain absolute clarity about individual responsibility. They can see exactly who processed each tube, who ran each assay, and who verified the results.

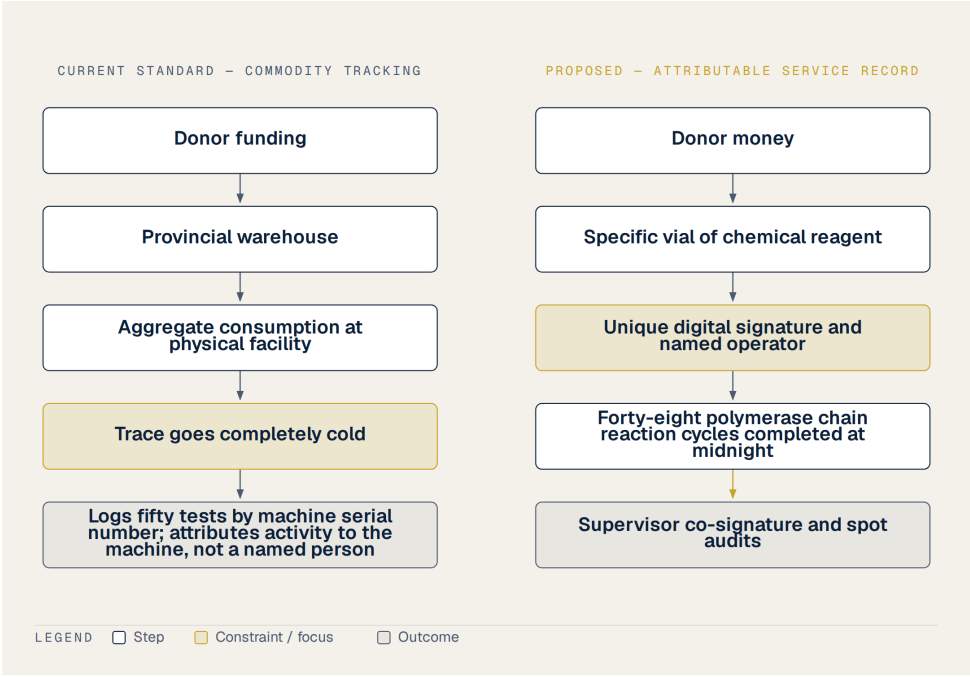
However, we must be completely honest about the limits of this design and preserve some important caveats. The ALCOA standard was originally created for highly regulated pharmaceutical manufacturing and clinical trial laboratories. None of these regulations were written for a provincial preventive medicine laboratory in Vietnam. Applying them here is a deliberate borrowing of high-status industry practices, not a legal requirement. Furthermore, in the existing industry frameworks, attribution is treated almost entirely as a mechanism of blame and auditing. It is designed to find out who to blame when a record is wrong or when a test fails. Shared logins are a classic inspection finding because they destroy this accountability. The breakthrough of our design is that we do not use attribution to police or blame our staff. Instead, we transform this institutional record of responsibility into a positive, portable professional credit owned by the worker. We make the same record serve both the technician and the institution without adding administrative weight.

In those years, I handled thousands of diagnostic test kits bought with international donor money. I saw how much effort went into tracking these resources. Yet, as I write this proposal, I must point out a massive gap in how we account for these materials. This brings us to the third reader of our proposed record: the funder.

Under our design, the donor finally gains the ability to see exactly when a reagent bought with its money was used, by whom it was used, and for what specific test.

To understand why this is a breakthrough, we must look at how donor commodities are tracked today. The global health system relies on major supply chain programmes and software platforms like OpenLMIS. These systems are supported by organizations like Gavi and the Global Fund. But if you look closely at these tools, they only trace commodities as far as the physical facility. They show that a box of reagents arrived at a provincial warehouse or was delivered to a specific clinic. After that, the trace goes completely cold. The systems can only record aggregate consumption. They cannot link a specific vial of chemical reagent to an individual diagnostic test. They cannot tell you which named operator actually ran the test, or on which day, or for which patient. (41,42)

Some people argue that modern instrument connectivity solves this problem. Many laboratories use middleware to connect high-throughput machines, such as viral load analyzers, directly to database systems. But this connectivity does not solve the human accountability gap. It attributes activity to a physical machine rather than a named person. The system logs that the analyzer ran fifty tests, but it does not tell you who prepared the samples, who calibrated the instrument, or who was responsible for the run.



Where the trace goes cold. Current tracking terminates at the clinic door and the machine casing. The proposal extends the same trace by one more step, to the person who ran the test.

It is worth being exact about what exists today and what does not. There is currently no open-source laboratory information management system in low and middle-income countries that features cryptographic immutability by default. No international donor currently requires tracing chemical reagents down to individual operators. Implementing this level of traceability is our own deliberate design choice, not a legal mandate or a donor requirement. The limits of input honesty set out earlier apply here in full: the chain protects a record after it is written, and nothing more.

By closing this gap, we transform how donors view their investments. Instead of just seeing shipments arrive at a building, they see their funding actively turn into verified human labor on the laboratory floor.

Much hard work goes completely unrecorded. The most significant gap in our entire global health data architecture. When you look at how laboratory data is managed today, every single existing framework treats attribution as a mechanism for blame and auditing. In these regulatory structures, knowing who performed an action is only useful when something goes wrong. If a data entry is incorrect, or if an assay fails, the system is used to identify the person responsible. It is a tool for policing and finding fault. Shared logins are treated as major inspection findings because they break this chain of blame. Yet, this focus on fault-finding wastes something.

Currently, no regulator, no laboratory information system, no logistics platform, and no international donor has turned this very same record into positive, portable credit owned by the person who performed the work. If a technician runs four thousand tests over three years with absolute precision, that achievement is completely lost to history. The system logs their name for auditing, but they leave their job with a blank paper resume. There is absolutely no established standard for exporting a verified service history as a professional asset. Under the current system, a worker's hard labor is treated as an institutional commodity rather than their own professional property. We have built elaborate, expensive structures to trace reagents and monitor machinery, but we have completely failed to build a mechanism that allows workers to own the proof of their own competence.

It is worth being plain about the existing tools and their limitations. Some administrative platforms, such as iHRIS or the Workload Indicators of Staffing Need tool, are designed to track staffing numbers, training levels, and overall workload estimates. However, these systems remain completely disconnected from actual daily laboratory logs. Other modern security frameworks, such as blockchain protocols for auditing electronic health records, are de-

signed to log who accessed or edited files. Yet, their focus is entirely on data security and detecting abuse. They were never intended to record positive professional achievements or provide portable career credits. Furthermore, we must acknowledge a vital caveat. No open-source laboratory database in low and middle-income countries currently features cryptographic immutability by default, and no international donor requires tracing work down to individual operators. Implementing this in our local clinics is a choice, not a legal requirement. By transforming the institutional audit trail into a verified service record, we can allow workers to own their professional history. This turns a tool of surveillance into a ladder of opportunity, helping the unconnected prove their worth.

When we talk about global health, we often speak in terms of charity or moral obligation. There is a version of this argument addressed to funders instead, and I will put it plainly rather than at length. Programmes that support laboratory capacity abroad do so on the reasoning that detection is cheaper and earlier at the source. That reasoning depends on the people at the source still being there next year.

This is the honest reading of what our proposed system is and what it is not. I must state a vital caveat from the outset. This recognition pathway is not an early warning system itself. It will not automatically detect a new pathogen or trigger an emergency alert. Instead, it is the quiet machinery that keeps the workforce operating the early warning system from falling apart. By transforming invisible daily sweat into a verified professional asset, we ensure that trained technicians stay in their jobs longer and produce more timely, reliable diagnostic data. We are securing the human foundation of health security. Without stable, motivated people on the laboratory floor, even the most expensive surveillance machinery is useless.

The cost of failing to stop health threats at their source is staggering. To understand the scale of this risk, we can look at the economic impact of the last major global crisis. According to a 2020 paper by Cutler and Summers published in the *Journal of the American Medical Association*, COVID-19 cost the United States roughly 16 trillion dollars. This is an almost unimaginable sum of money. However, we must preserve the strict honesty caveats attached to this figure. First, the text notes that about half of that 16 trillion dollar estimate represents non-market health loss, such as the value of lost lives and mental health impacts, rather than direct lost economic output. Second, we must acknowledge that this paper is a short viewpoint article rather than a peer-reviewed, complex economic modeling study. It is a simplified estimate, not a definitive final calculation (3,4).

Yet, even with these necessary caveats, the message is undeniable. The economic and human cost of a delayed response dwarfs any local investment in public health systems. When a provincial laboratory in a place like Vietnam cannot retain its staff, the delay in detecting a novel threat can ripple across oceans in days. By investing in a digital pathway that verifies and honors the work of frontline technicians, the United States keeps the people who operate that capacity in place. Equipment is the part that gets counted. The person who runs it at two in the morning is the part that walks away, and replacing her costs more than keeping her.

During that time, I watched valuable investments quietly decay. When we talk about global health security, we often look at the staggering scale of the financial numbers involved. If we look at the cost of global pandemic preparedness, the figures presented by major international health bodies are massive. The G20 High Level Independent Panel, in its July 2021 report, estimated that global pandemic preparedness requires roughly 15 billion dollars a year in ad-

ditional international funding. Just a few months later, in March 2022, the World Health Organization and the World Bank presented a report to the G20 estimating the total preparedness need at 31.1 billion dollars a year, with 10.5 billion of that representing additional international funding. On these numbers, the record deserves a plain statement. They are not bottom-up cost calculations with precise confidence intervals. Instead, they are expert consensus estimates designed primarily for policy advocacy (3,4,16).

Some advocates justify these investments by quoting dramatic return-on-investment multiples. The trouble is that the published multiples do not agree with one another; they differ by orders of magnitude depending on which costs are counted and over what horizon. In this proposal, I deliberately refuse to quote any specific return multiple. I do this because those multiples are simple ratios rather than counterfactual models. They do not survive strict economic scrutiny and fail to account for complex real-world variables.

However, the core of my argument does not rely on these grand global estimates. There is a deep cost asymmetry at play here. While the world debates tens of billions of dollars, the local reality on the laboratory floor is far more modest. This proposal does not ask for a single dollar of new, unallocated money. It does not ask for a new budget line from any donor or ministry. Instead, the real argument is that this proposal asks that what has already been invested does not decay (3,4).

In Vietnam, the marginal dollar should not be used to build new facilities or buy more high-throughput machinery. We already have a preventive health network that reaches every province and extends down to the commune level, though as of June 2025 there are thirty four provinces where there were sixty three, and the network is still settling into that new shape (21). The legal rails are already in place through Circular Thirty-Two, which mandates continuous

medical education hours. Even the US CDC has already invested in deploying open-source laboratory information systems across thirty-eight local laboratories. They have also supported quality management systems to help ten laboratories achieve ISO accreditation. The infrastructure is there, but without a stable, motivated workforce, it is slowly falling apart as trained technicians leave for the private sector. This proposal is incredibly cheap because it does not require us to construct something from nothing. It merely asks to connect the existing pieces that are already paid for, so that earlier investments keep working. It is a small, controlled option on a massive potential payoff.

During those years of processing diagnostic samples, I often thought about where these threats actually originate. In this proposal, I want to write about where new diseases actually come from, and I want to do so carefully, without pointing fingers or placing blame on local cultures. When a new virus emerges, it is not a moral failing of a specific community. The real scientific drivers of emerging infectious diseases are ecological and physical. They are driven by rapid land use change, rising livestock density, and the expanding human-animal interface.

The data on this is very clear. In a landmark 2008 study published in *Nature*, Kate Jones and her colleagues analyzed three hundred and thirty-five emerging infectious disease events. They found that sixty point three percent of these emerging infectious disease events are zoonotic, meaning they jump from animals to humans. Furthermore, seventy-one point eight percent of those zoonotic events originate specifically in wildlife. A later study in 2017 by Allen mapped these hotspots globally. His team showed that these hotspots are heavily correlated with land use change, mammal diversity, and human population density. Southeast Asia, including Vietnam, frequently appears on these maps. This is not because of

cultural habits or local eating practices. It is because our region has a high diversity of mammals, particularly bats, combined with rapid forest fragmentation and dense human and livestock populations (43–45).

But we must look at the structural forces behind these ecological pressures. It is necessary to be honest about the fact that forest fragmentation and agricultural expansion in developing nations are substantially driven by global economic demand for commodities. Additionally, we must state a critical scientific caveat regarding these hotspot maps. Hotspot maps are partly an artifact of surveillance effort. Countries that spend more money looking for viruses and reporting them will naturally show up as hotspots, whereas regions with minimal surveillance remain blank. The map reflects where we look, not just where the viruses are.

This is why Vietnam's position matters. We are a frontline partner and a gatekeeper in global health security. We stand at the precise interface where these ecological pressures collide. This is where the modest claim made earlier has to be repeated rather than escalated: the pathway does not detect anything. It keeps the people who operate the detection in place. The marginal dollar spent here by international partners should not be used for building new clinics or buying redundant equipment. We do not need more concrete. Instead, the marginal dollar must be spent on activating what already exists, connecting the legal rails and local infrastructure we have already paid for.

There is a standard objection at this point: that aid only works where institutions are already strong, so weak systems should be built before they are funded. That claim came from Burnside and Dollar, and it did not survive replication; Easterly, Levine and Roodman showed in 2004 that the interaction disappears once the dataset is extended. The related folk history, that Germany and

Japan recovered because their institutions outlived their factories, is also not what economic historians find. I raise these only to set them aside, because neither the objection nor its rebuttal decides anything here (29).

Therefore, institutional capacity does not determine whether money should flow. It determines the channel through which money works. In Vietnam, our marginal dollar does not need to build new capacity from scratch. We do not need a massive, expensive overhaul. The preventive medicine network is already established in every province down to the communes, Circular thirty-two already provides our legal continuing medical education rails, and the US CDC has already placed open-source laboratory information management systems in thirty-eight labs. Our proposal is incredibly cheap because it does not demand new capacity. It simply connects what already exists.

In writing this proposal, I want to discuss the official rationale of global health security and the honest limits of the statistics we use to support it. The overarching framework of our field is organized around the Global Health Security Agenda, which has used the prevent, detect, and respond triad since 2014. The United States published a Global Health Security Strategy in April 2024 that reinforced this framing. That strategy has since been withdrawn and was replaced in September 2025 by the America First Global Health Strategy, which moves to multi-year bilateral agreements meant to end what the State Department calls a prolonged period of dependency. I mention the change because a proposal that cites a withdrawn strategy as current policy deserves to be read with suspicion, and because the newer framing makes this argument easier rather than harder. Read the commitments in it. Those agreements are to guarantee that frontline healthcare workers keep being funded for the term, and the department undertakes to work with each

country to ensure there are data systems in place. Frontline workers and the systems that record them are named in the same paragraph of current American policy. What is missing is the join between them. A record that stays with the worker is exactly what survives a transition to self-reliance. Officially, the United States CDC's Division of Global Health Protection reports an impressive role in supporting the containment of over two hundred and fifty outbreaks in the year 2024 alone. This is the grand narrative that justifies international intervention. (46–48)

However, when we look at the specific performance targets on the ground, the numbers become much more normative than experimental. Consider the widely cited seven-one-seven target, which calls for detecting an outbreak within seven days, notifying public health authorities within one day, and completing an early response within seven days. This formula is useful, but we must state clearly that it is a normative target chosen for memorability rather than a precise threshold derived from scientific experiments. When we examine actual implementation data published in 2023 across forty-one health events in countries like Brazil, Ethiopia, Liberia, Nigeria, and Uganda, the reality is sobering. While fifty-four percent of events met the detection target and seventy-one percent met the notification target, only forty-nine percent met the early response target. In fact, only about twenty-seven percent, or roughly a quarter of all assessed events, met all three targets. (16,48)

The same pattern shows in those financing estimates I quoted earlier. They are expert consensus shaped by policy advocacy, not bottom-up costings with confidence intervals around them. That does not make them worthless; it makes them political indicators rather than blueprints, and a proposal that leans on them should say which of the two it is relying on. (3,4,16)

Our proposal is not trying to resolve these global macroeconomic debates. We are proposing a highly contained, local pilot of a verified service record for our technicians. By running a single staff position on our existing legal continuing medical education rails, we can test these behavioral models cheaply. If we want to achieve targets like seven-one-seven, we must first make the daily, invisible labor of our local laboratory technicians visible.

From Record to Door

In London a surgical trainee opens his intercollegiate e-logbook and enters a laparoscopic cholecystectomy. His supervisor countersigns it the same day. Three years of hernia repairs, appendix removals and assisted incisions sit in that file, each one dated, each one signed by somebody other than him. At national selection the panel does not ask him to describe his experience. It reads the count.

At the same hour in a provincial CDC in Vietnam, a laboratory technician finishes her ten-thousandth polymerase chain reaction assay. She has been at the bench since the previous evening. When she logs off, the count goes nowhere. There is no file with her name on it, no countersignature, nothing a committee could read. Her five years exist in shift rosters and internal reports that were never built to leave the building, and in the memory of a supervisor who will retire.

Two people, the same trade in the sense that both are trusted with a patient's diagnosis, the same hour of the same day. One of them has a record. That is the whole of the difference, and it is not a difference in skill.

During those long shifts of processing diagnostic assays, I watched how the absolute invisibility of our field work limits our professional lives. When we propose a digital registry to record the verified contributions of our technicians, we must look at the real-world models that actually work. The British surgical logbook, the

flight logs of the Federal Aviation Administration, and the portfolios of the National Board for Professional Teaching Standards in the United States all share the same structural engine. They succeed because they meet three rigid conditions. First, the record is verified by an external supervisor. Second, the record is tied directly to a real career gateway held by an authority. Third, the reward is substantial, such as a specialized training slot or a salary upgrade, rather than a symbolic badge.

The weakest link is worth naming first. Our technology can easily handle the verification of these records. However, the system only works if we can secure a real educational partner to hold open the career gateways. We are proposing tiered scholarships of twenty-five, fifty, and seventy-five percent for our certified technicians rather than symbolic micro-credentials. I should state the evidence position on badges once, carefully, because it is easy to overstate and I have overstated it in earlier drafts. I am not aware of independent peer-reviewed evidence that a digital badge, on its own, raises promotion or retention; most of the effectiveness material I could find comes from the platforms that issue and sell them. That is an absence of evidence in the sources available to me, which is not the same as proof that badges do nothing, and I would revise this paragraph if a reader pointed me to a trial I have missed. What follows from it is modest: do not build the pathway on the badge. Build it on the gate the badge is supposed to open.

Our daily work vanishes into thin air. We ran thousands of critical assays, but when we went home, we had nothing to show for it. In this proposal, I am suggesting that we build a verified portfolio of labor. To some, this might sound like a theoretical experiment, but we do not need to invent this concept from scratch. We can look directly at three concrete, highly successful models that already run on this exact architecture.

The first model is British. Surgical trainees in the United Kingdom do not apply with reference letters describing their promise. They submit the summary sheet of a consolidation report from the Intercollegiate Surgical eLogbook, and the NHS guidance is unusually specific about what it will accept. The sheet shows the number of procedures and the date range. Cases assisted or supervised count; cases merely observed do not. Every page must be signed by a consultant giving their full name, their General Medical Council number and the date, and that consultant may ask to see the whole logbook before signing. The resulting count is then banded: forty cases or more scores an A, thirty to thirty-nine a B, twenty to twenty-nine a C, eleven to nineteen a D, ten or fewer an E. (49)

That system settles several things before anyone argues about them. It defines what counts as taking part. It requires an attester who is himself identified by a registration number, so the countersignature can be traced to a person who can be asked about it. And it converts a count into a band, so two candidates can be compared without anyone writing a paragraph about character. None of this is novel and none of it is expensive. It is simply decided, and written down, and applied the same way to everybody.

The second model is found in aviation. After the Colgan Air crash in 2009, Congress directed the Federal Aviation Administration to raise the experience required of airline pilots. The rule that resulted, 14 CFR 61.159, sets an airline transport pilot certificate at fifteen hundred hours of logged flight time, and 14 CFR 121.436 is what obliges the airline to hire only pilots who hold it. A logbook is not a resume builder in that system. It is the legal gate. (50)

What matters more for this proposal is the part of the rule that is usually left out of the summary. The fifteen hundred hours are not absolute. A pilot who graduated from an approved four-year aviation degree can hold a restricted certificate at one thousand hours;

a two-year degree brings it to twelve hundred and fifty; a military pilot to seven hundred and fifty. The regulator did not lower its standard for those pilots. It accepted a different, verified record in place of raw hours. That is exactly the shape of what I am proposing, and it already exists in American federal law, in a safety system with no appetite for sentiment. A verified record of structured work can substitute for a cruder count, provided somebody outside the applicant certifies it. (49–51)

The third model belongs to education. American teachers can seek National Board Certification, and they do not do it by sitting an exam. They assemble a portfolio: video of their own classrooms, work their students actually produced, and written analysis of both. Twenty-nine states then attach money to the result. Washington pays 6,514 dollars a year, with up to another 5,000 for teachers in qualifying high-poverty schools. Hawaii pays 5,000 and doubles it for high-need schools. Delaware pays twelve percent of base salary. Idaho and South Dakota pay 2,000. (51)

Two things are worth taking from that spread. The first is that the money is real enough to change a household budget, which is what separates this from a badge. The second is that it is not guaranteed: in March 2025 Washington was reported to be considering cutting the bonus for thousands of its certified teachers. Even the strongest example of a record converting into money is a political decision that can be revisited, and a proposal that pretends otherwise is not being straight with anyone. (51)

These three diverse fields prove that a self-recorded log can become a powerful professional asset. They also agree on what makes one work, and the three conditions they share are worth taking one at a time.

The weakest link in this proposal is the gate. A verified ledger of laboratory work is worth little unless somebody on the other side of

a door agrees to read it. If nobody does, we are handing our staff symbolic digital badges.

I should be careful here, because in an earlier draft I wrote that a single educational partner was the whole prerequisite, and then spent several pages describing American certification routes and graduate admissions as though they were separate benefits. Both cannot be true. What is true is that there is more than one gate, and they are not equally likely. The nearest and most realistic is domestic: a provincial or national training institution, a promotion panel, a transfer decision, a place on a course. Further out, and for a small minority, sit the international routes described later in this chapter, which are real but narrow and slow. The proposal needs at least one gate open to be worth building, and it should be judged on the domestic one, since that is where almost every technician this book is about will actually stand. The evidence position on badges stated earlier applies here. I have to secure a real gatekeeper to make this work.

The first condition is that the record of labor must be verified by someone outside of the worker. A self-declared logbook holds no weight on its own. In a laboratory setting, a technician cannot simply write down that they ran a test and expect others to believe it. There must be an independent verifier, such as a laboratory supervisor, who confirms the entry using a unique digital signature. Without this external verification, the database is nothing more than a list of unproven claims.

The second condition is that the log must be tied to a real door that someone already controls. In the UK surgical model, that door is a national selection panel for training. In American aviation, it is the legal gate to an airline first officer seat. For teachers in the United States, a complex portfolio of classroom video and student work converts directly into a substantial annual stipend. If we build a

system of records without a corresponding gatekeeper who holds the keys to real career advancement, we are wasting our time. The record must lead directly to a specific, tangible opportunity that already exists in the administrative framework.

The third condition is that the reward must be substantive. By substantive, we mean actual training opportunities, professional licensing, or direct financial compensation, such as an annual stipend. We should not be offering our workers symbolic tokens or digital badges. The honest position on this point, caveat included, is this. This is the same limit noted above. Every single claim of their effectiveness traces back to the marketing materials of the platforms that sell and issue these badges. There is no trial data supporting their value on their own.

Our proposal is built on these three pillars. It is an honest design that avoids the trap of empty gamification. Underpaid staff deserve better than a sticker. Instead, we want to build a verified record that connects directly to the existing legal rails of continuous medical education and real training slots. By respecting these three strict conditions, we can turn a simple laboratory log into a genuine professional asset that our workers can carry with them to build their futures.

I know how easy it is to make a paper log look perfect. But when we talk about creating digital, immutable records of laboratory labor, we have to face a hard truth. A tamper-proof log of fabricated entries is still only a tamper-proof lie. Cryptography can prevent anyone from changing a record after it is saved, but it cannot stop a technician from entering fake data at the start. To solve this hardest of problems, we must build an objective verification layer. That layer is external quality assessment, commonly known as EQA.

Under EQA, the laboratory is sent blind proficiency panels without any warning. Because these samples arrive randomly and are

prepared by an external party, they cannot be self-reported, anticipated, or gamed. The technician must process them exactly like regular samples. EQA is not a speculative or experimental concept. It is globally recognized as the gold standard for laboratory quality. Participation in EQA is strictly mandatory under international standards like ISO 15189, which requires labs to join inter-laboratory comparison programs and document corrective actions when results drift. In the United States, passing proficiency testing under the CLIA rules is a mandatory condition of laboratory certification (52–54).

Vietnam is not starting from scratch here. We have a solid legal framework for this. Under Decision 2429 of 2017, issued by the Ministry of Health, laboratories are evaluated using a national quality criteria set. Our designated national quality control centres manage these schemes and publish national proficiency testing data. Furthermore, we have shown that we can push this high level of quality control deep into the field. The National Institute of Hygiene and Epidemiology and the Pasteur Institute have successfully taken external quality assessment down to the commune level for HIV testing. We have proven that even remote, rural health staff can participate in rigorous quality control networks (53,55,56).

However, we must be completely honest about the limits of these existing programs. Every EQA scheme running today is designed to assess the laboratory as a whole institution, not the individual worker. Currently, there is absolutely no published data on individual-level EQA performance for provincial CDC technicians in Vietnam. Suggesting that we already evaluate individual personnel through EQA here would be mere speculation. Furthermore, there is a real risk of purpose drift. Historically, under CLIA, laboratories have been caught giving special treatment to proficiency samples because the stakes were too high. When you turn EQA into a high-

stakes test of an individual, you create a blame culture that forces people to hide errors rather than report them. To avoid this trap, our proposal does not record whether an individual passed or failed a panel. Instead, we log that a technician has successfully participated in a set number of external quality rounds under a rotation mechanism, alongside their documented corrective actions when discrepancies occurred. This is meant to record professional discipline without turning quality assessment into a game of fear, but I do not want to claim it escapes the problem it is designed around. It relocates the problem. If a logged corrective action feeds a scholarship tier, then a discrepancy becomes something a technician and their supervisor both have reason to bury, or to paper over with a corrective action written for the file rather than for the bench. The incentive lands on the same pair of people who already sign off on everything else.

There is also a volume problem worth conceding. An external quality panel arrives a few times a year. It cannot testify to the thousands of routine assays described at the start of this book; it can only testify that when the laboratory was tested blind, it performed. Those are different claims. External assessment is what makes the record credible to an outside reader, and routine volume is what makes it substantial to the worker who built it. Neither one substitutes for the other, and a pilot that measures only the first will not know whether the second is honest. No system in the world has yet turned EQA participation into a portable individual credential, but by doing so, we can finally verify the worker's true competence.

Our staff struggle to prove their daily labor. In this proposal, I want to address a critical design correction regarding how we verify laboratory labor through external quality assessment, or EQA. Standard practice in global health always scores the laboratory as a

whole institution, not the individual person. If you look at standard laboratory audits, this institutional focus seems absolute. However, there are some narrow, existing precedents for individual tracking that we must examine. Under the American CLIA rules, proficiency testing samples must be handled by the routine staff who actually perform the testing, and these samples must be rotated among them. This rotation means the specific individual running the sample is already documented in the laboratory's records, and this proficiency testing can serve as a valid input for annual personnel competency assessments. Furthermore, the British NHS cervical screening programme goes even further and actively assesses individual staff competence, covering around one thousand one hundred and nineteen personnel across sixty-three laboratories twice a year, while schemes like UK NEQAS Digital Morphology also accept individual registrations.

Therefore, our proposed system makes a vital design correction. The digital ledger will never show a personal pass or fail score for an individual technician. Instead, the database will document two things only. It will show a technician's active participation across official sample rotations. It will also record their documented corrective action when a result was out of consensus with the reference group. Logging how a technician handles a discrepancy shows true professional discipline and honesty. It proves they can participate in a rigorous quality loop without turning quality assessment into a high-stakes test of fear. This design correction ensures that we protect data integrity while respecting the human limits of our front-line workforce.

However, we must examine what the leading standards authorities say about using external quality assessment as a direct measure of an individual's skill. The United Kingdom Accreditation Service, or UKAS, released a position paper in 2017 stating clearly that ex-

ternal quality assessment is not a compulsory component of assessing individual competence. Instead, EQA is designed to guarantee the overall quality of the service being delivered, not to act as a personal test. In agreement with this view, the Clinical and Laboratory Standards Institute, or CLSI, frames competence as a multi-element assessment. In their QMS03 guideline, they explain that true competency evaluation must combine direct observation, result reviews, blind specimens, and problem-solving exercises. Relying on a single test score is inadequate. (28,57)

The core of the trap lies in how we evaluate accuracy. When a monitoring system scores performance based on peer consensus, and the entire peer group begins to drift in the same direction, the accurate laboratory is suddenly the one that looks wrong. In this scenario, the person who reports truthfully becomes the outlier. Instead of being recognized for their precision, they are treated as an error that must be corrected. This is not a theoretical suspicion. Vietnam's own published proficiency testing data across fifty-nine laboratories, published by Van and colleagues in 2023, shows a clear divergence. In their study of nine biochemical analytes across twelve samples, the expert group consensus actually diverged from the overall participant consensus on common analytes. This means that peer group divergence is a measured, documented phenomenon and not just a vague suspicion. It proves that peer groups can and do drift, and that the crowd is not always right. (10,57)

If we build a registry that punishes the outlier, we will receive uniform reports, but we will not receive true ones. Technicians are rational actors. If telling the truth exposes them to administrative correction or shame, they will simply adjust their numbers to match the consensus. This defeats the entire purpose of public health surveillance.

To avoid this, our proposed registry implements two critical design consequences. First, the record must be designed as an asset owned by the worker rather than an audit instrument held by the institution. Under a worker-owned model, honest recording builds your own professional capital and establishes your personal credibility. Under an institutional audit model, honest recording merely exposes you to punishment. Second, we must ensure that scoring references an objective laboratory method rather than a fluctuating crowd wherever it is possible. This anchors our evaluations in science rather than peer consensus.

During those years of processing diagnostic samples under constant pressure, I dreamed of a way to make our labor visible. But as I write this proposal, I must state a vital, hard boundary before anyone else points it out. It is worth being plain about what our proposed verified service record cannot do. A verified digital work history, no matter how securely recorded on a digital ledger, cannot substitute for formal professional licensure or national board certification.

The first gate is professional certification. The American Society for Clinical Pathology Board of Certification runs a domestic credential, MLS(ASCP), and an international one, MLS(ASCPi). The domestic credential has six eligibility routes and I checked them on the Board's own page rather than trust a summary, because the summaries get this wrong. Route 1 wants a United States accredited laboratory programme. Route 2 wants an existing MLT(ASCP) credential, which a technician in Vinh Long does not have and cannot easily get. Route 3 was discontinued on the first of January 2023. What is actually open to a foreign-trained applicant is Route 5, and Route 5 does not start in the United States at all. It requires the international credential first, then a transcript evaluation verifying equivalency to a United States baccalaureate, then five years of full-

time clinical experience within the last ten, across blood banking, chemistry, haematology, microbiology, immunology and urinalysis. (58)

That last clause is the argument of this book, stated by somebody else. The five years must be documented on the Board's own Experience Documentation Form, signed by a laboratory director. The work has to have happened, and the only proof the Board will accept is a form filled in afterwards by a supervisor who has to remember it. A technician who did the work and whose director has retired, or whose institution has merged into another province, has no route at all. Not because she lacks the experience. Because she cannot produce the paper. (21,58)

The second gate is state licensing. About eleven states in the United States, including California and New York, license laboratory personnel separately from national certification. These states conduct their own audits, reviewing academic coursework hour by hour against their own rigid category lists. Because foreign transcripts are structured differently, they often fall short in a single chemistry or biology category. When this happens, the licensing board does not care about years of experience. The only accepted remedy is to go back to school and complete the missing coursework.

The third gate is the visa, which is completely independent and is almost always the hardest barrier to cross.

The honest conclusion is that our proposed digital record changes none of these rigid international rules. It does not waive a single coursework hour or bypass a visa interview. However, it does something that is worth far more than it first sounds. It makes the clinical experience route provable. To qualify for certification through experience, the board requires detailed proof of acceptable work in an accredited laboratory, verified directly by a laboratory

director. Today, a technician from a provincial CDC in Vietnam has no credible way to prove their years of invisible labor to an international board. By providing a tamper-proof, verified history of their actual diagnostic contributions, we turn their silent service into a verifiable asset. It does not guarantee a visa, but it gives a hard-working technician a real fighting chance.

We must not make promises to our workforce that we cannot keep. This proposal does not offer a magic key to escape across borders. Instead, we must present this system for what it actually is: a tool to make the “N years of experience” route provable. It is a verified asset for postgraduate admissions, fellowship applications, and research roles where formal licenses are not required. By acknowledging these institutional barriers plainly, we protect our integrity and build a system that remains useful within its true boundaries.

To understand why this record matters, we must look at global migration. The data is sobering. A June 2021 report by Batalova and Fix from the Migration Policy Institute analyzed the American labor force. They found roughly eight million college-educated immigrants. But one in four of them, or two million people, is unemployed or working a job that only requires a high school education. This is brain waste. Those educated abroad are underemployed at a markedly higher rate of twenty-nine percent, compared to twenty-one percent for those educated domestically. English proficiency is the strongest predictor of this gap. But the lack of verifiable work history is a silent, massive barrier (54,57,59).

Our proposed record directly chips away at this barrier. It does not replace the exam. But it makes the existing experience route provable. The Board accepts clinical experience in place of an American laboratory programme, but only when it is attested on its own form by a laboratory director. Our secure, tamper-proof ledger provides that provable history. Beyond boards, the record carries

weight in graduate admissions. Most admissions committees now use holistic reviews. In fact, roughly sixty percent of Master of Public Health programs no longer require standardized GRE tests. What a verified service history can do in such a file is narrower than I first wrote: it substantiates a claim the applicant is already making about their bench experience, rather than outranking transcripts or referees. I have no evidence about how admissions committees weigh an unaccredited foreign logbook, and I should not pretend otherwise. It opens doors. It provides undeniable proof of competence for fellowship applications and research roles that do not require state licenses.

Yet, we must be completely honest about our goals. Crossing an ocean is not the priority for most of our technicians. For them, the largest realistic market for recognition is inside Vietnam itself. We want to solve a domestic problem. By establishing a portable, verified record, we allow underrepresented local technicians to prove their worth. They can move across provincial boundaries. They can apply for national programs or seek donor-funded research roles. We do not promise an easy visa or a guaranteed salary bump. There is no trial evidence for those outcomes. We simply provide a verified asset. It lets our frontline workers own their own labor.

Here in Vietnam, our public health institutes have already delivered quality systems at scale. The commune-level quality assessment work described earlier is the proof that this reaches rural staff, not only provincial ones. However, I must state an important fact that we cannot afford to gloss over in this proposal. No per-technologist external quality assessment data exists publicly in Vietnam. Our national data is published entirely at the laboratory level, meaning we have no public record of how individual technicians perform on these tests.

There is an older way of allocating professional opportunity in any public service, and it is worth naming plainly rather than implying it. A training slot, a transfer, a place on a course: these have often moved through who knows whom, who spoke for you, whose department you came up in. Our proposed system of verified service records runs on something else entirely, namely what you did. Both approaches share the same ultimate objective, which is to prove that a worker is trustworthy and capable. However, one of these approaches allocates professional opportunities by connection, while the other allocates those exact same opportunities by verified evidence. In sociological terms, this is a shift from particularism to universalism. It shifts the entire workforce from a model of personal patronage to one of objective, standardized proof.

Despite these academic debates, the practical lesson for our healthcare system is clear. By building a verified digital record of service, we create an honest ladder for the unconnected. This proposal does not ask to change national civil service laws or demand new funding lines. It simply turns invisible daily labor into a portable, verified professional asset. It allows the quiet work done in provincial labs to speak for itself, bypassing the need for personal connections and ensuring that merit, not patronage, determines who rises.

I must state our limits honestly. There is currently no trial evidence or randomized controlled study evaluating how this decentralized, worker-owned model affects reporting accuracy or staff retention in low-resource provincial laboratories like ours in Vietnam. We are proposing a logical, evidence-informed framework to test these ideas. By giving our technicians ownership of their verified logs and anchoring their scores in objective methods, we can turn our registry into a safe ladder for professional growth rather than a dangerous trap.

Plugging Into Rails That Already Exist

In a small, humid laboratory in a remote district of a developing country, a molecular diagnostic machine hums softly as it runs. Its network cable stretches across a concrete floor to a router, transmitting data in real time to a polished dashboard monitored in a European capital. The database registers its serial number, its calibration status, the lot number of the reagents, and the precise temperature profile of each polymerase chain reaction. Beside the machine stands a young technician who stayed past midnight to load the samples, but her name is entirely absent from the digital feed. The dashboard sees the instrument, but it does not see her. Three countries away, inside a darkened storage room of another provincial hospital, a second identical machine sits sealed in its original shipping crate. Delivered two years ago under a major international aid grant, it has never been unpacked, connected to a power outlet, or loaded with a single specimen. Because it is offline, it sends no telemetry and generates no records, remaining completely invisible on the donor's digital dashboard. It is invisible in exactly the way that matters most, existing only as a quiet line on a physical inventory sheet while patients wait outside.

Our current tracking systems, including major platform networks that collect data across forty-three countries, view attribution purely as an instrument of surveillance and auditing. They know which serial number ran which test. Whether they know who

ran it is a more careful question, and this chapter answers it inconsistently in places, so let me settle it here. Instrument software and middleware usually do record an operator account. What that account establishes is that someone signed in as that account, which is a weaker claim than it looks in a laboratory where logins are shared during a surge, and it is a claim that never leaves the institution's own server. So the gap is not that the machine records nothing. It is that what it records is neither reliably tied to a person nor readable by anyone outside the building. In Vietnam, thirty-eight laboratories have implemented open-source laboratory information systems with support from the US CDC. These provincial systems mostly lack audit trails that bind a run to an identified person in a way an outside reader could check.

Our proposal is to add the missing layer to this running infrastructure by connecting a machine's serial number to a technician's name. Under our system, this data becomes a portable, verifiable professional asset owned by the worker rather than an auditing tool held by the institution. To do this cheaply and securely without expensive blockchain technology, we propose a Postgres append-only database. Each daily log of events is connected in a hash chain and anchored to a public ledger using a trusted RFC 3161 timestamp and OpenTimestamps, which are free to use and will outlive our organization. This ensures the records cannot be retroactively altered when an institution closes (60).

An unalterable record of something that never happened is still a lie, and now a permanent one. Cryptography alone cannot prevent data gaming or falsified inputs. No trial evidence exists on how a worker-owned, machine-anchored registry affects reporting accuracy or retention in low-resource settings. That absence covers everything proposed in this chapter, and I will not restate it at each step. We are proposing a logical, evidence-informed twelve-month pilot.

Linking the machine's serial number to a name turns anonymous telemetry into something a person can carry. I should be exact about what that does and does not establish, because it is the first thing a reviewer will press on. A serial number tied to a login proves that a particular account started a particular run at a particular minute. It does not prove that the specimen was collected properly, that the standard operating procedure was followed, or that the result was correct. Those are separate questions, answered by the external quality assessment layer and the supervisor countersignature, not by the telemetry. What the telemetry establishes is narrower and still worth having: that the work happened, when it happened, and who was logged in when it did.

I watched our daily work disappear as soon as the samples were processed. When writing a proposal to improve staff retention in preventive health, people often assume you must ask for new regulations or complex bureaucratic reforms. But we do not need to write a new law. We already have a powerful, legally binding rail running through our medical system, Circular 32/2023/TT-BYT, issued by the Ministry of Health.

Under Circular 32/2023/TT-BYT, every practice license holder in Vietnam is legally required to complete at least one hundred and twenty credit hours of continuing medical education every five years. Obtaining this CME certificate is a mandatory condition for license renewal. The administrative rail is already laid down. Every technician in our provincial laboratories is already forced to run on it (61).

However, there is a massive gap in how this legal rail functions today. The current system counts only passive hours attended. It simply measures how long a technician sat in a classroom chair. This metric is completely disconnected from actual field performance. A specialist can run thousands of critical diagnostic assays

during a pandemic outbreak, but none of that invisible daily sweat counts toward their educational credits. As far as I have been able to establish, it also connects to no structured path to promotion, though I am reporting the absence of a link I could not find rather than a rule that forbids one. This disconnect is the precise seam where a verified professional record fits. By converting real-world labor into verified, credit-bearing assets, we can align legal compliance with actual competence.

This integration is especially vital for preventive medicine staff. Unlike clinical workers in hospitals, CDC technicians do not have private clinical revenue streams, making them the hardest to retain in the public sector. While the government has tried temporary salary bonuses, global evidence suggests that financial incentives alone represent the weakest intervention for long-term retention. Underutilized institutional capacity, like our existing CME rail, is much cheaper to activate than constructing new systems from scratch.

Still, we must remain completely honest about the limits of our proposal. There is currently no published evaluation of dedicated career pathways or retention outcomes for CDC and preventive health workers in Vietnam. Globally, the World Health Organization recommends linking continuing education to career advancement, but they openly acknowledge that the evidence backing these guidelines is of low or very low quality. Furthermore, using educational scholarships instead of cash rewards to incentivize this credit rail has not been documented in the literature. The absence of trial evidence noted earlier applies to this link too. Indeed, we must admit that providing training without corresponding increases in salary or responsibility can backfire, causing even greater frustration and demotivation. We are proposing a logical, low-cost connection of existing pieces, not a proven cure. By inserting a verified

record into this existing legal seam, we can make the labor of our frontline technicians visible without asking for a single new regulatory decree. (6,12)

When international partners talk about improving global health security, they often propose building massive new systems. They want to construct new clinical facilities or buy expensive, high-throughput machinery. But this proposal is different. This is not a proposal to build something new from scratch. Instead, it is a proposal to connect things that are already standing. We do not need new infrastructure, because a significant foundation of technology and quality standards is already in place.

We should look at the investments that already exist. As documented by Landgraf and his colleagues in a 2016 study, the United States CDC successfully deployed open source laboratory information systems to thirty-eight laboratories across Vietnam. This was not a temporary donation. These systems were deployed with a clear plan, and the financing and ongoing maintenance have been transitioning directly to Vietnamese institutions. In addition to software, the US CDC supported quality management systems and the Strengthening Laboratory Management Toward Accreditation program. This systematic effort brought ten key laboratories to international accreditation, including our national reference laboratories for HIV and tuberculosis (52,53).

This international accreditation is tied directly to our own national standards. The Ministry of Health grades laboratories on a five level scale. Under this official framework, the top two levels, four and five, require international ISO 15189 accreditation. This proves that the legal rails and the quality benchmarks are already written into our administrative system. The pieces of the puzzle are lying on the table, paid for and operational (52,53).

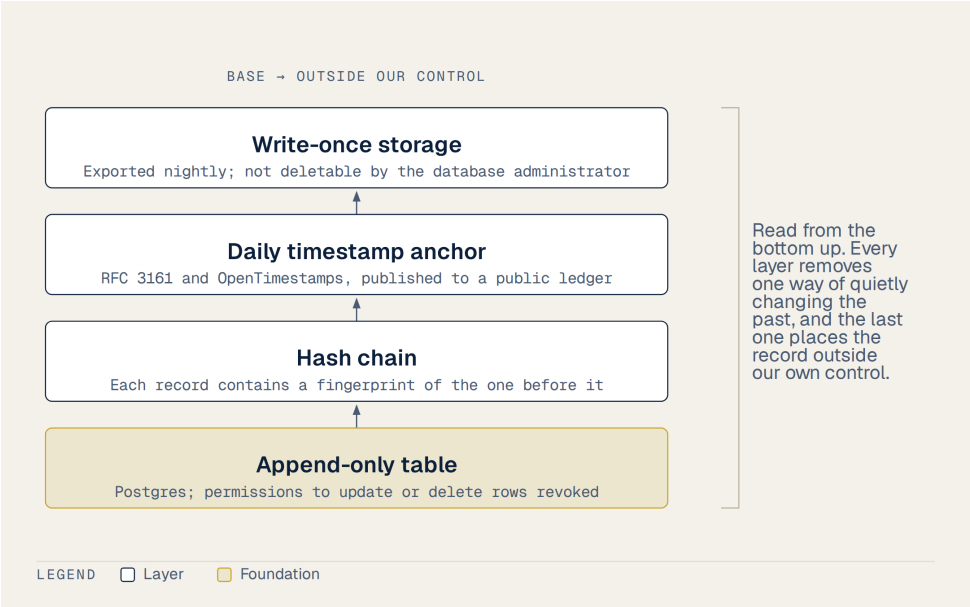
However, we must be completely honest about the real state of these systems. If we look beyond the well-funded central reference laboratories, the reality in provincial and preventive labs is highly pieced together. While the central facilities enjoy robust laboratory information systems, our provincial CDC labs often operate with fragmented, local solutions. Crucially, we must state a hard scientific caveat. There is currently absolutely no published evidence of LIMS audit trails achieving ALCOA plus standards at the provincial level in Vietnam. Attribution in the sense used here, an identified person rather than an account, is what is missing. The connection between individual labor and institutional quality remains broken.

Our proposal does not ask for new buildings or redundant equipment. We do not need more concrete. We simply ask to activate this underutilized institutional capacity. By linking our proposed verified service record to the existing open source information systems and the Ministry's quality grading scale, we can make the daily labor of provincial technicians visible. We are connecting the existing LIMS deployments, the ISO accreditation requirements, and the local workforce. This connects the standing pieces of a long-term international investment, transforming a routine laboratory log into a portable professional asset.

I have learned to distrust systems that promise to solve everything with one switch. This is why the proposal keeps the technical layer deliberately modest. When people hear about secure, immutable digital ledgers, they immediately think of complex, expensive infrastructure. We do not need that. We can build an exceptionally secure system using simple, well-established technologies that cost almost nothing.

Our proposed architecture relies on four basic blocks. First, we use a simple database with an append-only table. By revoking the administrative permissions to update or delete rows, we ensure that

new data can only be added to the bottom of the ledger. Second, we connect these database rows using a hash chain. In this chain, each new record contains a mathematical fingerprint of the record that came immediately before it. This mathematical link binds the entire history together. If anyone tries to modify a past entry, the entire chain of fingerprints breaks. Third, a hash chain alone is not enough because the database operator could still rewrite the entire history from scratch. To prevent this, we anchor the head of the chain daily. We use a trusted timestamp and a free, public timestamping service to publish the latest hash. This anchor places the record completely outside of our control. Fourth, every night, we export the ledger to write-once-read-many storage. Once written to this storage, the files cannot be deleted or altered by anyone, not even the database administrator.



Four cheap blocks, no blockchain. Each layer is ordinary and already in use somewhere else. It is the stacking that does the work, not any single one of them.

We must explain plainly why this design does not use blockchain. For years, blockchain has been promoted as the ultimate solution for medical records. But we must face the reality of the scientific literature. After years of attempts, blockchain for health records remains stuck entirely at the pilot stage. These pilot programs routinely run into major, unresolved barriers regarding interoperability, scale, cost, and governance. Furthermore, almost every serious blockchain proposal actually keeps the medical records off-chain. This means the blockchain is only used to store access logs or file hashes. A simple hash chain and a public time-stamp service achieve this exact same security function without any of the massive overhead. Increasingly, international donors and funders read the word blockchain itself as a major credibility risk. They view it as a technological red flag.

When we propose to record this invisible work on a secure database, the next logical question is how we present this data to the outside world. We do not want to create a closed system that only we can read. We need an outward-facing credential that can be easily shared and verified across borders. Fortunately, we do not have to design a new standard for this. We can build on top of a powerful, globally recognized credential architecture that was finalized in June 2024 (49–51).

This standard is Open Badges version three. In 2024, the 1EdTech consortium officially approved this updated framework, defining it strictly according to the W3C Verifiable Credentials standard. By adopting this technology, we can issue a periodic verified service record to our laboratory technicians. This record states that a specific worker has performed a specific number of verified assays during a given period. We sign this record digitally using our organization's private cryptographic key. Because it conforms to the W3C standard, anyone, including a university admissions commit-

tee or a scholarship board, can verify the signature instantly. They do not need to email us, and they do not need to contact the issuing laboratory. The credential proves itself mathematically in a matter of seconds. Still, we must remain practical. While version three is the modern standard, version two remains far more common in actual daily use. To ensure maximum compatibility, our system should issue both formats in parallel (22,23).

We need to show our funders that this is not a private, speculative invention. This is precisely where global health security is already heading. The World Health Organization already operates a live public key infrastructure for digital health certificates. This system, known as the Global Digital Health Certification Network and SMART Trust, is already functioning at an immense scale. For instance, in 2024, it was used to verify digital health cards for more than two hundred and fifty thousand pilgrims during the Hajj. More importantly, the published roadmap for this global network explicitly includes a plan to certify public health professionals through the WHO Academy (23).

Our proposal merely aligns our local laboratory records with these massive, pre-existing global rails. We are not asking our international partners to trust a custom, unproven technology. We are simply adopting the exact same cryptographic standards that the World Health Organization is using to secure the global health workforce. By doing so, we ensure that a technician from a provincial laboratory in Vietnam can carry their verified service history anywhere in the world as a trusted, portable asset. They can prove their competency to any partner, bypassing the need for personal connections. This simple technical layer turns invisible daily labor into a globally readable truth. (22,23,62)

I watched so many of my colleagues' applications go completely unnoticed. When a provincial technician tries to apply for a post-

graduate program or an international fellowship, their file hits a wall. The admissions committee in the United States or Europe looks at a folder from an unknown clinic in a remote province. They have no practical way to verify any of the claims in that file. If they try to send an email to our facility, it takes weeks to route. Often, it is never answered. Even when an answer does return, the committee has no frame of reference to assess whether the reply is authentic or what the facility's standards actually mean. Under these conditions I would expect a busy admissions officer to set the application aside, though I am describing what the incentives point to rather than any documented admissions behaviour. It is not done out of malice, but out of administrative self-defense.

A verifiable credential changes this equation, and it is worth walking through exactly what the committee does, because an earlier draft of this chapter described it in a way that cannot work. I wrote that the committee opens the ledger, types a code and reads the history. That would require a public database of technicians' daily work, which is precisely the thing this chapter spends several pages refusing to build. Both statements cannot stand, so let me replace the wrong one.

The worker presents the credential; nobody looks it up. The record itself lives in a store the worker controls, and when she applies for a place on a course, she sends the committee a signed document, in the same sense that she would attach a transcript. The committee does three things with it, none of which involve browsing anything. It checks that the signature belongs to the issuing laboratory, using a key the laboratory has published. It checks that the document has not been altered, by recomputing its hash and finding that same hash committed to the public timestamp layer at the date claimed. And it checks that the credential has not since been withdrawn, against a revocation list the issuer maintains.

What the public layer holds is only the commitment: a salted hash and a time. It says that this exact document existed on that date and has not changed since. It does not say who the technician is, what she tested, or how many samples she ran, and it cannot be browsed to find out. The distinction matters because it is what lets the same design be both checkable by a stranger and closed to one.

The speed of the lookup is not the part that matters. The work is done by two words: partner laboratory. An admissions committee at a major foreign university has never heard of a provincial CDC in Vietnam. They have no reason to trust our local records. However, they do know and deeply trust the international institutions that sponsor us. They know the United States CDC, and they know the global public health bodies that vouch for our system. The temptation here is to say that the committee's trust in that institution transfers to a technician they have never heard of. It does not, and I should not write as though software could accomplish something that only governance does. An admissions board does not admit a stranger because a serial number checked out. Trust in a person still comes from an accredited institution supervising them and a named human being willing to put their name to a letter about them.

What the cryptography does is narrower and still worth having. It removes one specific reason to discard an application: the suspicion that the claimed experience was invented. When a technician writes that she ran ten thousand assays over five years, the committee currently has no way to test that sentence and no cheap way to ask. A verified record makes the sentence checkable in a minute. It does not make her a good candidate. It makes her a candidate whose claims are worth reading. Everything after that is the ordinary business of references, transcripts and interviews, and none of it is displaced by a hash.

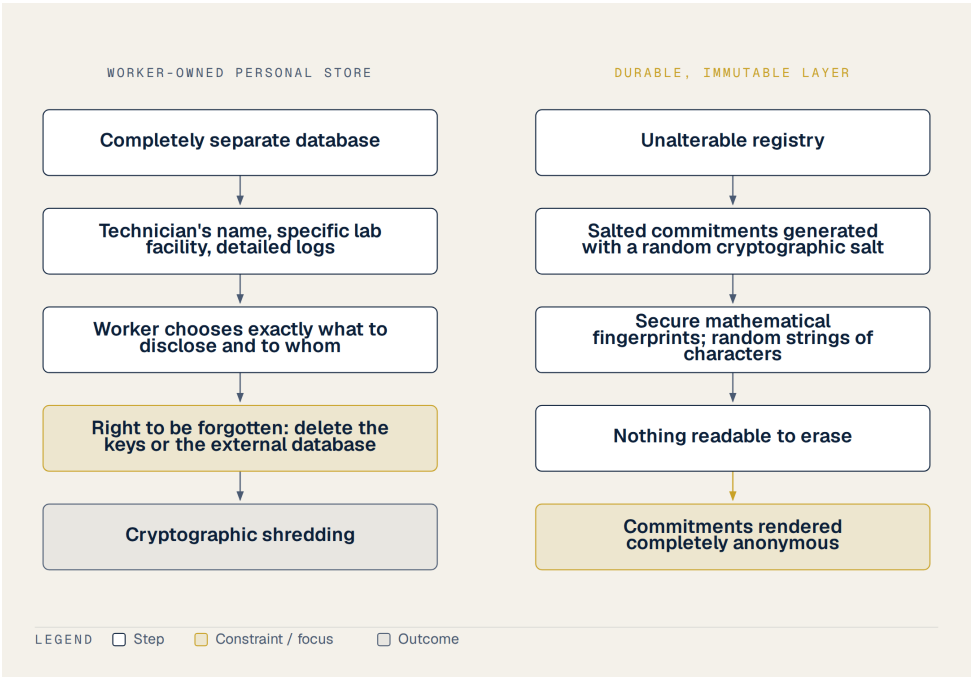
On what this transfer of trust can achieve, the record deserves a plain statement. There is no trial evidence proving that a lookup identifier directly increases acceptance rates for Vietnamese technicians. We cannot guarantee that this system will magically solve the complex visa barriers that represent the hardest gate for international migration. We also do not have any data showing how many international programs will actively accept these verified credentials instead of requiring traditional academic transcripts. What we do know is that it solves the immediate, silent barrier of verification cost. It allows a local technician's actual work to speak for itself. By linking our daily labor to the existing reputation of our global partners, we turn a provincial log into an internationally readable asset, giving our frontline workers a fair chance to be seen.

That leaves the question of who is protected from the record. When we create an immutable digital ledger to verify our service records, we must face a major structural challenge. How do we protect the privacy of the laboratory technicians themselves? This is the core reason why our proposed ledger is not a public database that anyone can browse to spy on a worker's daily habits. If we simply made a public search engine for laboratory actions, we would be placing highly sensitive personal data into an immutable layer. This approach would collide directly with international data protection laws.

To understand this conflict, we must look at the official guidelines issued by the European Data Protection Board. Their guidelines on blockchain and durable ledgers, finalized in July 2026, state a clear and absolute principle. The board explains that technical impossibility is no excuse for failing to erase personal data under privacy laws like the right to be forgotten. Under these strict rules, even placing a hash or an encrypted version of personal data on an unalterable registry can represent a serious compliance viola-

tion. Once personal data is written to an immutable chain, it can never be truly deleted, which means the system is legally broken from day one (60,63).

Therefore, our proposal implements a vital privacy design correction. We separate the unalterable registry from the actual personal details. The technician must always remain the sole owner of their data. Instead of publishing records openly, the worker chooses exactly what to disclose and to whom. This selective disclosure is precisely what verifiable credentials were built to achieve under the W3C standards. On the durable, immutable layer, we write only salted commitments. These are secure mathematical fingerprints that contain a random cryptographic salt. To anyone browsing the public registry, these commitments look like meaningless, random strings of characters. The actual personal data, including the technician's name, their specific lab facility, and their detailed logs, is stored in a completely separate database. Because this personal store is separate, it can still be deleted if the worker requests it. If a technician decides to exercise their right to be forgotten, we simply delete the keys or the external database. This action immediately renders the unalterable commitments completely anonymous, a process known as cryptographic shredding.



Erasable person, permanent proof. Deleting the personal store breaks the link. The chain survives intact and stops pointing at anybody.

While this privacy architecture is theoretically sound and aligns with the best practices of the European Data Protection Board, its long-term stability and compliance record in a provincial Vietnamese setting remain untested, per the caveat entered at the start. We are proposing a logical, standard-compliant path, but we cannot claim it has been tested under real-world clinical audits. By keeping the personal data completely off-chain and giving the technician full control over selective disclosure, we protect our frontline workforce from surveillance while building a system that respects international law.

In this proposal, I want to address the fundamental problem of how our professional competency is evaluated. This is a classic economic challenge. As the economist George Akerlof showed in his

work on market pooling, when competence cannot be observed from the outside, the market simply pools everyone together. They price all workers at the average. Under this pooling system, the highly skilled technicians are discounted alongside the weak. We have no way to separate ourselves or prove our actual worth to international partners. A credible signal, as Michael Spence demonstrated in 1973, is the only mechanism that can break this pooling (62,64,65).

However, building such a signal requires scale. This is a classic club good, an idea explored by Buchanan in 1965. A register containing only one laboratory is completely useless, whereas a register that coordinates five hundred laboratories becomes a standard that every committee and admissions board is forced to read. The value of the network only exists when we cross that threshold of scale (62,64,65).

Fortunately, we can look at successful precedents that have already run this race. The first is ORCID, a persistent identifier system for researchers that anyone can join. It successfully gathers verified contributions, peer reviews, and publications into a digital portfolio that researchers carry for life. The second is GISAID, a pathogen genome platform. Its core mechanism is built on the rule that sharing data earns the contributor professional credit. Yet, GISAID also teaches us a much harder lesson about governance, showing the intense conflicts that arise over who holds the keys to the club and who controls access to shared data.

This brings us to the cold start problem, a challenge we must state explicitly. A network is virtually worthless until it is populated with a critical mass of active users, meaning the very first technicians who join must bear all the administrative costs of participation without receiving any immediate benefit. Because of this cold start, we cannot simply launch a global club on day one. Instead, we

must start inside a group that already has established members. We can begin within an existing training programme alumni network, such as the Field Epidemiology Training Programme graduates, or we can restrict our initial rollout to a single, motivated province.

We must be entirely explicit that governance kills far more digital systems than technology ever does. To succeed, we must settle the political question before we write a single line of code. The data must belong entirely to the worker, and the organization only keeps the register. By establishing this clear rule, we ensure that our system is a neutral ledger of evidence, not a centralized surveillance tool.

Dozens of electronic reporting systems fail because nobody asked how they would survive the politics of ownership or the passage of time. When we propose a shared register to record the achievements of our local technicians, we must answer two existential questions before anyone even asks them. These two questions will decide whether our shared register lives or dies.

The first question is simple: who holds the keys to this data? The answer is that the data must belong to the individual worker, and the organization should only keep the register itself. GISAID, described earlier as a precedent worth learning from, is also the cautionary half of that lesson: its governance and access disputes are as instructive as its success. They show that when a central entity holds absolute control over access, the trust of the contributors quickly evaporates. To prevent this, our design ensures that the worker owns their personal record and controls who sees it through selective disclosure, meaning they can share their verified achievements without exposing their entire history.

The second question is what happens when the issuer of the record disappears. If a provincial health department or a local clinic shuts down or stops maintaining their server, the records they is-

sued must not vanish. The anchoring of these records must physically outlive the organization that created them. This is why our system uses a trusted timestamp and a free public anchoring service, rather than relying on a private, centralized database. Combining an RFC 3161 timestamp with OpenTimestamps anchors the commitment on a public ledger at no cost, so that part of the proof survives the organisation. I should be exact about how much survives, because the answer is less than the sentence above implies. What outlives the clinic without any further effort is the timestamp: anyone can still show that this document existed unaltered on that date. What does not automatically outlive it is the ability to say who vouched for it, because that depends on the issuing laboratory's public key still being resolvable. If the department closes and its key material is left to rot, verification degrades from "this laboratory certified this work on this date" to "this document is old and unaltered", which is a much weaker claim. Any serious deployment therefore has to say in advance where issuer keys are archived and who is responsible for keeping them resolvable after a closure. That is an institutional commitment, not a cryptographic one, and it is the sort of detail that decides whether a record is still worth anything in fifteen years (60). We do not need a private database or an expensive blockchain to achieve this.

The limits are worth stating plainly. This is a contained, twelve-month pilot, and, as stated at the outset, no trial has evaluated whether a decentralised record of this kind affects retention anywhere. We are not proposing a magic wand. But by answering who holds the keys and ensuring the records survive the organizations that issue them, we can build a portable ladder of opportunity for our frontline staff.

What Could Go Wrong

The air in the office is thick with the smell of wet concrete and ozone from the air conditioner. It is four o'clock on a humid Tuesday afternoon, and three of us are gathered around a wooden desk in our provincial CDC laboratory in Vietnam. We are reviewing the latest round of proficiency testing results before we submit them to the national quality center. On the desk lies a printed sheet with rows of figures representing the biochemical analytes from our molecular and analyzer runs. Under the national proficiency testing model, our results will be compared against the peer consensus of fifty-nine other laboratories across the country.

I look down at the tenth column on the paper. Our machine measured a value of one hundred and forty-two. But the preliminary reports from the other provincial units are already circulating on our phones, and everyone else has reported between one hundred and thirty-five and one hundred and thirty-seven. Our figure sits quietly outside that range. It is a clear outlier.

There is a long, heavy pause in the room. Nobody speaks. My supervisor taps his pen against his palm, a rhythmic, dry click that seems to grow louder with every second. He looks at me, then looks back at the sheet. Nobody says anything for a moment. Under a system where quality is audited through peer consensus, being different is dangerous. If we submit this number, our laboratory will be flagged. We will be subjected to administrative reviews, corrective action plans, and quiet blame.

The safest thing to do is very simple. We do not need to accuse anyone. We do not need to make a scene. We just need to look at the sample again. My supervisor gently suggests that we should check the calibration log once more, or perhaps rerun the assay, until our number finally agrees with the crowd. We all know what that means. If we keep looking, we will find a reason to adjust the number. We will align our column with the others so that our laboratory remains safe within the consensus.

This is the silent pressure of the peer consensus trap. We are reviewing nine common analytes across twelve samples, much like the published proficiency testing data compiled by Van and colleagues in 2023. In their study of fifty-nine Vietnamese laboratories, they documented how the overall participant consensus can drift, even diverging from expert reference values. One statement of the evidence position covers this whole chapter, so I will make it here and not repeat it at every step. Nothing proposed below has been tested in a trial. There is no published data on individual-level proficiency testing in Vietnamese provincial laboratories, no evaluation of layered administrative verification against data fraud in that setting, and no trial of point systems or worker-owned registries on laboratory retention anywhere comparable. Every design choice that follows rests on reasoning from adjacent fields and on what the failure modes appear to be, which is a weaker footing than evidence and should be read as such. But in this room, on this rainy afternoon, we do not discuss the literature. We simply stare at the tenth column. I pick up the pipettor. We prepare to look at the sample again, waiting for the number to agree, while the pressure in the room slowly settles over us like the humid air outside. (10,55)

During those long shifts of processing diagnostic samples, I learned that security is never just a matter of code. In this proposal, I must address the most significant technical vulnerability of our

entire system. It is a threat that cryptography cannot touch. We are proposing a secure, immutable ledger to record our daily laboratory service history, but we must admit a hard truth. An unalterable registry only guarantees that data has not been changed after it was written. It does not guarantee that the data was true in the first place. A tamper-proof log of fabricated entries is nothing more than a tamper-proof lie. Immutability defends against alteration. It does not defend against false input.

If we ignore this boundary, our advanced cryptographic architecture becomes nothing but expensive theatre. If a technician wants to inflate their work history, they can easily log fictitious assays or claim to have run samples that never existed. The database will happily secure these entries, chain them together with mathematical fingerprints, and anchor them to a trusted timestamp. The resulting credential will be mathematically flawless, yet completely fraudulent. We must not build a high-tech facade that secures empty, fabricated claims.

To prevent this exploitation, we must design verification layers that operate at the physical entry point. The answer does not lie in more cryptography. Instead, we must rely on practical administrative gates that ground our digital entries in physical reality, and they are the same gates named in the layered defence at the end of this chapter rather than a separate list. First, we must implement randomized spot audits. Under this mechanism, independent inspectors physically verify paper laboratory logs against the digital database. Second, every log entry requires a supervisor countersignature. I have to deal with the obvious objection here, because I raised it myself at the start of this chapter: in the scene I described, the supervisor is the person suggesting the number be adjusted. A countersignature from someone who wants the same result is worth nothing. That is precisely why it is the fourth layer and not the first,

and why it sits alongside a check the supervisor cannot influence. Contemporaneous capture happens before he sees the figure. The blind proficiency panel is scored outside the building by people who do not know whose result it is. The countersignature adds local accountability on top of those, and it fails in a different way than they do, which is the only reason to have it. Where it does help is the ordinary case, and the ordinary case is most of them: a second pair of eyes on the daily work, providing the external verification that any credible ledger requires. We can look at global precedents like the British surgical e-logbook or the FAA pilot flight logs. These successful models only function because they require external verification by a supervisor or a governing authority. Third, we can use simple, automated checks to flag impossible travel. For example, if our system receives a GPS log from a provincial outbreak site and a physical laboratory check-in from the same technician within the same hour, it must automatically reject both entries (49–51).

Those measures are ordinary administrative hygiene, and none of them has been tested against this problem. None of it has been tested against this problem, per the evidence position stated at the start. We are proposing a practical, low-cost defense system to protect our database's integrity, but we must evaluate its effectiveness under real-world clinical audits. Without these physical validation layers, our secure database remains completely vulnerable to the threat of fabricated entries. Cryptography without physical verification secures the wrong thing.

What we measure defines how we behave. When we design a system to track and reward performance, we face a well-documented risk. If we make the rewards too high or the indicators too rigid, workers will naturally find ways to manipulate the numbers. The indicators cease to represent quality, and instead become the target itself.

We have clear global evidence of this breakdown. Under performance-based financing schemes in Burkina Faso, healthcare providers falsified registers at scale. Their intense fixation on meeting specific indicators completely displaced the actual clinical objectives those indicators were meant to represent. The paper records became immaculate, but the actual patient care did not improve. We saw the exact same pathology in the English National Health Service under their strict target regimes. To meet administrative deadlines, providers engaged in documented clock stopping and selective referral. They manipulated the timing of admissions and chose patients who were easier to treat just to satisfy the official metrics. This is what happens when we turn a professional mission into a numbers game.

To prevent this systemic failure, we must follow three consensus design rules established by literature. First, we must keep the indicator set small. When we introduce too many measures, we do not get better data; instead, we simply create more unintended consequences and administrative overhead. Second, we must mix process measures with outcome measures. Tracking only outcomes encourages providers to select easier cases, while tracking only process metrics leads to empty compliance. Third, we must carefully size any incentive. The reward must be large enough to motivate our provincial technicians, but it must never be large enough to make systematic gaming rational. We want to encourage honest logging, not creative accounting.

The three rules are borrowed from a literature that never studied a provincial laboratory. This too is reasoning rather than evidence. We do not have randomized controlled trials showing the exact threshold where an incentive becomes too large and triggers fraud. Furthermore, we must acknowledge that when we link verified performance logs to career opportunities like scholarships, we are in-

roducing a powerful motivator. Even with the best design rules, some level of metric displacement remains a constant risk in any system that measures human labor. We must accept this limitation, run regular audits, and keep our expectations realistic. By keeping our technical indicators minimal and rewarding the act of participation rather than speed, we can protect the integrity of our front-line workers.

There is a failure mode here that looks exactly like success. Administrators love to offer training. It is cheap, visible, and easy to document. However, the World Health Organization's 2018 guideline for community health workers and evidence from programs in Ethiopia, Kenya, Malawi, and Mozambique show a clear pattern. Upgrading professional training while keeping a worker's actual responsibilities and salary completely unchanged is actively demotivating. It creates deep frustration. When a professional recognition pathway ends in a beautiful paper certificate and nothing else, it is actually worse than doing nothing at all. Still, we must remain completely honest about the strength of this evidence. The World Health Organization openly acknowledges that its global guidelines on health worker career progression are based on research of low or very low quality, meaning we lack robust clinical trials in this space (30,66,67).

We face a parallel warning from the field of quality assurance and patient safety. In clinical laboratories, we rely on External Quality Assessment, or EQA, to verify our accuracy. But if we use this performance data punitively to rank or punish individuals, we trigger a dangerous behavioral response. The literature on just culture, such as Boysen's 2013 research and patient safety studies in 2025, shows that a culture of blame drives professionals to hide errors rather than report and correct them. We have hard legal precedents of this purpose drift. Under the United States CLIA regula-

tions, laboratories are legally required to treat proficiency testing samples exactly like patient samples. This law was written because laboratories began treating these test samples with extreme favoritism to secure high scores. High-stakes grading corrupts the metrics we trust (54,57,68).

To prevent this corruption, our database does not record a personal pass or fail score for individual technicians. That would make the record a tool for surveillance and blame. As noted at the start, no published data exists on EQA at the individual technician level in Vietnam, so any application of personal EQA grading is highly speculative. Instead, we follow a supportive design. The record simply shows that the technician successfully participated in a specific number of external quality assessment rounds under a rotating schedule. If a discrepancy occurred, it documents the formal corrective action they took to resolve it. Documenting honest participation and structured corrective action demonstrates true professional discipline. It encourages transparency instead of fear. By avoiding the dual traps of empty certification and punitive grading, we can build a registry that truly supports our frontline workers. Our technical design must respect these human realities, or the entire system will fail.

In this proposal, we must look closely at how we design the rules for recognizing our technicians' work. Gamifying work is popular, but consumer research warns us of two major design traps that we must actively reject.

The first trap is status tier demotion, which was documented in marketing research by Wagner, Hennig-Thurau, and Rudolph in 2009, as well as by Drèze and Nunes. Their studies show that demoting someone from an established status level is incredibly dangerous. It leaves the demoted individual with lower loyalty and motivation than someone who never held any status at all. In our CDC,

if we strip a technician of their hard-earned professional tier because of a slow month or a temporary drop in sample volume, we will actively destroy their remaining dedication. To avoid this severe behavioral backlash, our system must ensure that once a technician earns a tier, there is either no demotion at all or a clearly signposted recovery window where they can easily restore their standing (30,67).

The second trap is point expiry pressure, which was explored in retail research by Taylor and Neslin in 2005. Forcing accumulated points to expire is coercion-adjacent. It does not motivate genuine, high-quality work; instead, it reliably drives data falsification. When our laboratory staff face a ticking clock that threatens to erase their accumulated record of service, their immediate survival instinct is to game the system to protect their past efforts. This pressure shifts their focus from quality diagnostics to empty compliance. To prevent this systemic failure, we propose that once a technician earns a point for a verified lab action, those points do not expire. They remain a permanent, unalterable part of their professional identity (14,67,69).

These design choices rest on less than I would like. The studies we rely on were conducted on commercial consumers in developed countries, not healthcare workers in low-resource settings, so we face a real translation risk. No trial has measured how point or status systems affect retention among laboratory technicians in developing countries like Vietnam. Furthermore, a major systematic review of gamification by Sailer and Homner in 2020 shows that while cognitive effects can be moderate, the behavioural effects are much weaker, at a Hedges' g of about 0.25. We also have no independent evidence that digital badges or points translate into official salary increases or career promotion under our current civil service rules. We are borrowing these concepts from commercial market-

ing to help technicians see their own progress, but we must not overpromise. By rejecting demotion and point expiration, we protect our staff from administrative traps while respecting their dignity (22,23,27).

One rule about data underlies all of this: immutability is not correctness. When you design a system that cannot be altered, you must design for this reality from the very first day. Genuine human errors will always happen in a busy lab. A technician will inevitably mistype a sample number or log the wrong reagent batch. On an immutable digital ledger, that error can never be edited away. We cannot simply erase our mistakes because doing so would destroy the mathematical audit trail.

Instead, we must solve this using the same logic that accountants have used for centuries. When an accountant makes an error in a ledger, they do not white-out the mistake. They post a reversing entry. Our proposed system uses an append-only architecture where corrections are simply appended as new entries. The new entry contains the correction and supersedes the earlier one, but both remain fully visible in the permanent log. This ensures that the history remains transparent and unbroken, while the current active view shows the corrected state.

However, this absolute permanence creates a direct collision with privacy laws, specifically the right to erasure. Under international frameworks like the European Data Protection Board's guidelines on blockchain and durable ledgers, finalized in July 2026, technical impossibility is no excuse for failing to erase personal data. If a technician leaves the profession and demands that their history be deleted, an unalterable database would normally violate this law. Putting encrypted or hashed personal details directly on an immutable chain is a compliance violation because those hashes can never be removed from the public ledger (60,63).

We resolve this collision by keeping personal data completely separate from the durable layer. We do not write names, employee IDs, or location coordinates directly into our immutable ledger. Instead, we store all personal identifying details in a traditional, deletable database. On the durable, immutable chain, we only write salted commitments. These are secure cryptographic hashes containing a random salt that looks like noise. If a worker requests the deletion of their records, we simply delete their profile from the external, deletable store. This action immediately renders the salted commitment on the immutable chain completely unlinkable, a process known as cryptographic shredding.

PERSON → AFTER ERASURE

Salted commitment rendered completely unlinkable

The mathematics survives; the person does not



Durable, immutable chain

Append-only ledger of salted commitments, a permanent audit trail



Random salt, then cryptographic shredding

Worker requests deletion; the profile is removed and the link breaks



Traditional, deletable database

Names, employee identifiers, location coordinates

LEGEND ☐ Layer ☒ Foundation

How a permanent record can still be forgotten. The chain keeps its shape. What it points to can be taken away.

Whether that satisfies a right to erasure in law is genuinely unsettled, and I should not claim otherwise: regulators have not agreed whether a salted commitment left on a public ledger still counts as personal data once the payload is destroyed. The design is built to be defensible on that point, not proven on it. Whether it works in a provincial CDC is a separate question, and unanswered. This privacy-by-design approach is highly logical, but it has not been tested under a real-world clinical or legal audit in our setting.

By separating personal data from the unalterable log, we can build a system that respects both data integrity and the law.

I watched our staff work through long nights under intense pressure. When you design a system that records daily laboratory actions to help these frontline workers progress, you must establish an unyielding ethical boundary. In our proposal, we define this boundary as a bright line test that anyone can easily apply. If a technician fully understood how the system gathers, processes, and verifies their data, and would still choose to join, and could leave at any time carrying their data and losing absolutely nothing, then the system is scaffolding. It is a genuine ladder for professional growth. However, if the system only functions because the staff do not understand how it works behind the scenes, or because leaving the platform costs them their accumulated history, then it is a dark pattern. It is an administrative trap designed to exploit them.

When we propose a digital registry to record the field contributions of our technicians, we must establish strict ethical rules written as hard regulations rather than soft sentiment. Each of these rules must be directly traceable to a documented failure in modern workplace technology. We are not designing a game to manipulate our staff; we are building a protective scaffold to respect their dignity.

The first rule is that every score or metric must be completely legible and contestable. Algorithmic management research shows that the core harm of digital tracking is a metric the worker cannot see, audit, or dispute. If our technicians cannot trace how their points are calculated, the system becomes a tool of quiet oppression. The second rule is that the institution must never use these points to set base pay, build rosters, decide promotion, or open a disciplinary file. That prohibition is about who may use the record and for what, and it needs stating carefully, because this book also

proposes that the same record open scholarships, which is plainly a career benefit. The two are consistent only if the direction of use is fixed. The record travels outward when the worker chooses to present it, to a scholarship board, a training panel, a future employer. It does not travel sideways into her own management's hands as an instrument for ranking her against colleagues or building a case against her. The moment her employer can pull the ledger to justify a roster or a reprimand, it stops being voluntary and begins to function as coercive surveillance.

The third rule guarantees data ownership. The worker owns and can export their complete data history, while the provincial organization sees only the high-level aggregates that the worker explicitly chooses to share. The fourth rule is that nobody is ever ranked against anybody else. Systematic reviews among community health workers show that public leaderboards destroy team collaboration and cause deep anxiety among those at the bottom. It is necessary to ban public comparisons entirely. The fifth rule is that exiting the system must be a single, frictionless action that costs nothing already earned. The sixth rule explicitly bans, by name, digital nagging, guilt tactics on a broken streak, confirmshaming, and obstructed exit paths. If a technician misses a week of entries due to an exhausting night shift, the system must not punish or shame them.

The seventh rule is that rewards must attach strictly to documentation and participation, and never to clinical outcomes or processing speed. Rewarding clinical outcomes or speed invites both data gaming and dangerous patient selection, as seen in historical gaming failures within the British NHS. The eighth rule is that we must never escalate reward sizes to buy artificial uptake. A classic workplace wellness trial published in the *Quarterly Journal of Economics* in 2019 found that larger incentives mostly ended up paying

those who were already highly engaged, which skewed the sample and failed to change behavior. Finally, the ninth rule requires that a qualified human sits in every consequential decision, with all system overrides transparently logged. (38,69)

These conditions are drawn from settings very unlike ours. The principles are well documented in the international labour and behavioural literature, but no trial has evaluated how they apply to preventive health workers in low-resource Vietnamese settings. We are proposing a logical, evidence-informed framework. By writing these ethical laws into our software, we ensure our registry remains a ladder of opportunity rather than a tool of exploitation.

A single error is rarely what breaks a system. When we analyze how public health monitoring fails, we must look at the framework that holds our entire design together. This is the Swiss cheese model of accidents. In this model, every layer of defense has holes in it, much like a stack of sliced cheese. A single error usually passes through one hole without causing any harm. Disaster happens only when the holes in successive layers happen to line up perfectly, allowing a hazard to strike. The most important thing this model tells us is not that somebody made a mistake. Instead, it shows how everyone involved can believe they are doing the right thing, while the system still carries them toward a fatal outcome. This is exactly the trap we see when an entire peer group drifts together, and the only accurate laboratory becomes the one that looks wrong because it sits outside the consensus.

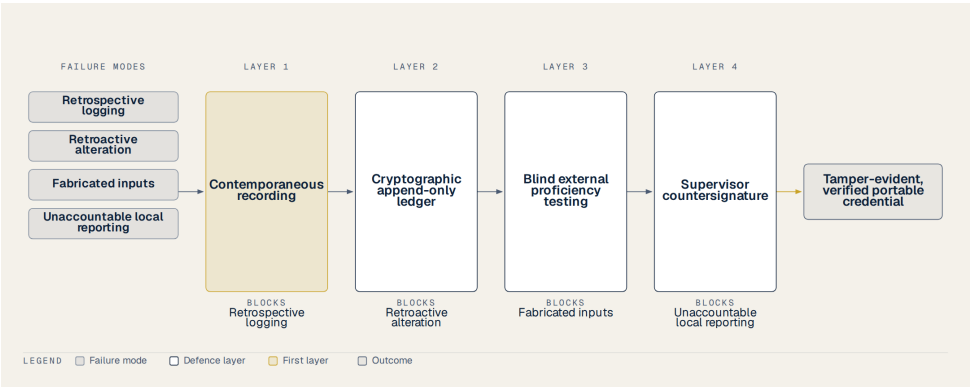
The corresponding discipline from industrial safety is to isolate a hazard and then verify that isolation rather than merely assuming it. This is the exact same instinct as sending a blind, unannounced sample to a laboratory to test its accuracy, rather than simply asking the operators how they are doing. External quality assessment and proficiency testing must act as this blind, independent check.

What follows for this proposal is clear. No single layer of defense is being offered as sufficient on its own. Instead, we propose a series of independent layers. Recording data at the exact moment of work is the first layer, preventing the common practice of retrospective logging. Immutability, secured through cryptographic append-only ledgers, is the second layer. Recorded history cannot then be altered after the fact. External verification through blind proficiency testing is the third layer. I have to be careful about what it catches, because an earlier draft of this chapter said it prevents the fabrication of inputs, and that is a category error I would rather correct than defend. A proficiency panel arrives a few times a year with a handful of coded specimens. It establishes that this laboratory, tested without knowing it was being tested, could actually perform the assay. It says nothing whatever about whether fifty runs logged on a Tuesday morning ever happened. A technician can pass every panel for five years and still invent a day's work, and no amount of blind testing would notice.

What the third layer defends against, then, is analysis that cannot be trusted, which is a real failure and a different one. Fabricated volume is defended against elsewhere, and it is worth saying exactly where, because this is the thinnest seam in the whole design. Three things bear on it. The instrument's own telemetry is generated by the machine rather than claimed by the operator, so a run that never occurred leaves no cycle record. Consumables reconcile against runs: fifty reactions consume fifty reactions' worth of reagent and fifty tubes, and a laboratory that logs work it did not do will show a stock ledger that does not balance. And randomised spot audit compares the digital record against the physical worksheets and specimen receipts for a date chosen after the fact.

None of the three is strong alone. Telemetry can be run on empty tubes. Stock records can be adjusted by the same people. Spot au-

dits are only as independent as the auditor. Their value is that defeating all three at once requires coordination across the laboratory, the store and the inspectorate, which is a materially harder thing to arrange than typing a number into a form. That is the honest claim: fabrication is made expensive and conspicuous, not impossible. A supervisor countersignature is the fourth layer, establishing clear local accountability. The value of this design lies in these layers being completely independent of one another, rather than in any single layer being exceptionally strong. An unalterable record of something that never happened is still a lie, and now a permanent one.



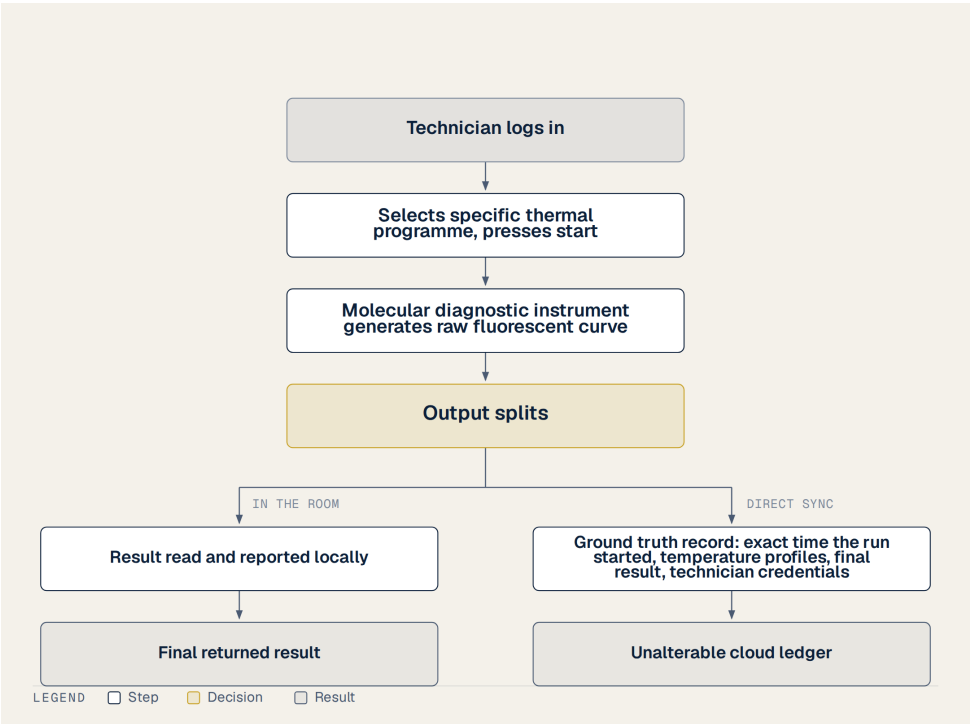
Four independent layers, four different failures. Cryptography alone stops exactly one of the four. That is the argument for layers, and also the argument against trusting any single one of them.

One last boundary. The registry as a whole remains untested for its effect on reporting accuracy or staff retention in low-resource public health laboratories. Furthermore, no system in the world has ever successfully converted individual-level proficiency testing into a portable, personal professional credential. We are proposing a logical, evidence-informed twelve-month pilot to test these behavioral and technical designs. By piling these independent, imperfect

layers on top of one another, we can build a secure pathway that keeps our frontline workers safe and our health security intact.

A Small Pilot, and an Invitation

On an ordinary Tuesday afternoon inside a single partner laboratory, a single molecular diagnostic instrument begins to run. A technician logs in. Her unique credentials are recorded. She selects the specific thermal programme for a polymerase chain reaction assay and presses start. Two hours later, a raw fluorescent curve appears on the screen. Each of these small facts immediately leaves the room by itself. They do not wait for anyone to write a report. They do not wait for an administrative sign-off. The exact time the run started, the temperature profiles, and the final result are synchronized directly to the cloud. This data flows instantly, without passing through any human hand that could edit, delete, or adjust the files. The machine simply accounts for its own work. This is the one part of the proposed system that nobody can quietly adjust to meet an institutional target, and it is worth being precise about what that buys, because it is easy to inflate. The record is trustworthy about a narrow set of facts: that a run began, at that minute, on that machine, under that account. It is not ground truth about competence. The instrument does not know whether the master mix was made correctly, whether the specimen matched the form, or whether the person logged in was the person at the bench. Everything to do with whether the work was done well is established elsewhere in this design, or not at all.



The record leaves before anyone can tidy it. No administrative sign-off, no report delay, and no window in which the record can be adjusted.

During those long shifts of processing assays, I watched how easily large software initiatives collapse. When we outline our first step, we must keep it plain and small. It must not be a laboratory information system, which has a soft boundary that anyone can argue already exists. In fact, thirty-eight of our laboratories in Vietnam have already implemented open-source LIMS with support from the US CDC. However, these existing platforms lack machine-enforced audit trails, and they are managed provincially without strict individual accountability. Trying to deploy another massive software suite will only drag us into endless bureaucratic debates.

Instead, our proposed pilot focuses strictly on telemetry from a single connected instrument at a single partner unit. The question

of whether artificial intelligence should monitor or even operate such a system is real, but that discussion belongs strictly to a later phase. A first pilot must be small enough that failing is cheap. This is a contained experiment. We are proposing a single position, a single twelve-month term, utilizing existing training rails, and demanding no new national budget lines. If the pilot fails, the cost is bounded, and the technical lesson is documented. If it works, the mission gains a baseline of objective truth that nobody has ever built. What matters is the principle for choosing an instrument, not which machine gets named. We should choose the instrument the funder already paid for. By doing this, the funder who bought the machine receives the machine's own account of how it was used, while the local operator receives evidence that she was the account under which it ran.

That is deliberately weaker than saying she gets credit for the work, and the difference is where a reviewer should push. In a laboratory where a senior technician's session stays open through a shift while junior staff do the extraction and pipetting, the telemetry credits the login, not the hands. Attaching scholarships to that number without addressing it would invite exactly the behaviour it rewards: staying logged in. So the instrument record cannot be the unit of credit on its own. It has to be paired with the countersignature, where a supervisor attests that this named person did this work, and with the requirement that accounts are individual and sessions time out. Telemetry then does what it is good at, which is fixing a time that nobody can move afterwards, and the human attestation carries the part telemetry cannot reach. A pilot that measures throughput without measuring how often logins are shared will not have tested the thing that matters. One device can thus serve all three readers at once without extra friction. A polymerase chain reaction platform is the obvious example of such an instru-

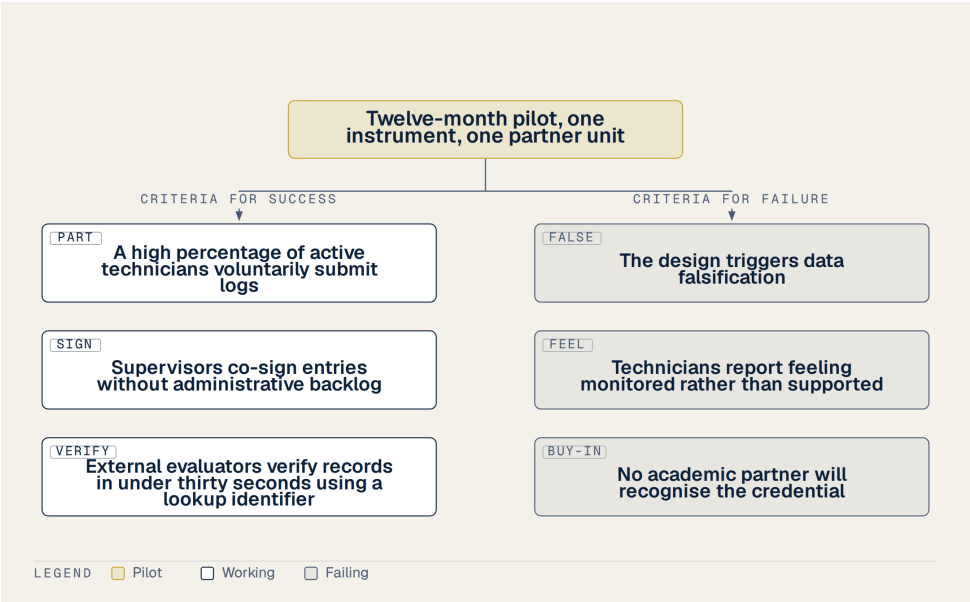
ment rather than a rigid requirement. It sits exactly where this mission touches the ground. Molecular diagnosis is the core of early outbreak detection. Fortunately, these PCR machines are already physically in place from the pandemic.

This is the honest reading of our limits. No trial has evaluated how a direct telemetry model of this kind affects reporting accuracy or staff retention, and that absence applies to every proposal in this chapter rather than to this paragraph alone in low-resource public health laboratories. We are proposing a logical, evidence-informed design to test these ideas in our local context. By starting with one machine and one connection, we can prove our concept without risking our scarce resources on a sprawling, unmanageable failure.

The most beautiful ideas on paper can easily die of institutional friction. To prove that a digital performance register can truly work in our health system, we must not launch a grand, sweeping national program on day one. Instead, we must propose a highly contained, twelve-month pilot project described with utmost modesty. We need a quiet, low-stakes experiment that operates within one established training programme, such as our Field Epidemiology Training Programme alumni network, or restricted to a single province. By keeping the boundaries of our pilot small, we ensure that a potential failure will only teach us valuable lessons cheaply rather than destroying public or political trust.

To make this pilot concrete, we must focus on a very small, practical set of indicators. We should not attempt to track dozens of complex clinical outcomes. Instead, we can record basic participation metrics, such as the successful logging of diagnostic assays and the documented completion of rotating external quality assessment rounds. More importantly, we must explicitly state on day one what counts as success and what counts as failure. Two different things move through this system and the pilot has to measure both, be-

cause an earlier draft blurred them. The instrument’s telemetry is automatic: it leaves the machine whether anyone wants it to or not, and nobody submits it. The professional record built on top of it is voluntary, because the technician decides whether to claim a run as her own work, and her supervisor decides whether to countersign that claim. So the pilot counts as working if a high proportion of active technicians choose to claim and keep their records, if supervisors countersign without backlog, and if an external evaluator presented with a credential can verify it quickly. Conversely, the system fails if it triggers data falsification, if technicians report feeling monitored rather than supported, or if we cannot establish active partnerships with academic institutions to recognize these credentials.



What would count as the pilot working, and what would count as failure. Both columns were written before the pilot starts. That is the point of writing them down.

Furthermore, we must agree in advance on who reviews the results. We cannot grade our own homework. We should establish an independent oversight committee before the first entry is ever written to our ledger. This committee should consist of independent public health experts, representatives from our international sponsoring institutions, and local medical audit authorities who hold no personal stake in the system's success.

On what we are proposing, the record deserves a plain statement. Nor has anyone measured career tracking against retention in Vietnam's preventive medicine sector. This twelve-month pilot is a contained, low-cost experiment. It runs on existing legal rails like our five-year Continuing Medical Education requirements, meaning it demands no new national budget line or sweeping legislative reforms. If the pilot fails, the financial and institutional loss is strictly bounded, and we will document the breakdown transparently. If it succeeds, we will have built a proven, scalable ladder that turns invisible daily labor into a portable professional asset for our frontline workers.

Honesty is our only real currency. In writing this proposal, I have to set out the limits of the evidence we rely on. We are proposing a system to recognize and retain frontline health workers, but we cannot hide behind technical optimism. We must state our boundaries clearly. To begin with, the global guidelines issued by the World Health Organization regarding health worker retention rest on shaky ground. The organization openly admits that almost all of its own recommendations in this critical area are based on low or very low certainty evidence. The comprehensive Cochrane review conducted by Grobler and colleagues in 2015 arrived at the exact same conclusion. There are simply no high-quality randomized controlled trials to prove which interventions actually keep workers at their posts (7,8).

This lack of comparative research is particularly true for our core incentive design. In our proposal, we choose to offer tiered scholarships rather than traditional cash bonuses, but we must admit a hard reality. There is no meta-analysis anywhere in the global literature that directly compares scholarships against cash bonuses for frontline health workers. We are choosing this path because we believe educational opportunities provide a more sustainable, non-coercive ladder, but this choice is a logical hypothesis rather than a proven fact.

Furthermore, we must look honestly at the history of compulsory service commitments. Many governments try to solve the rural retention crisis by forcing health workers into mandatory service contracts in exchange for training or support. However, the evidence tells us this approach is deeply flawed. A comprehensive review of compulsory service programmes across seventy countries, published by Frehywot and colleagues in 2010, found that most of these programmes were never rigorously evaluated. The review demonstrated that while mandatory service contracts succeed in temporarily filling vacant posts during the period of legal obligation, they lose these people almost immediately after the obligation ends. Forcing people to stay simply creates an unhappy, transient workforce. The research shows that what actually predicts long-term retention is working conditions, equipment that functions, and salaries paid on time (70,71).

Therefore, we must completely reject the idea of using our professional register as a lock. We should never frame a service commitment as a legal trap to bind a technician to a remote clinic. Instead, our system must be designed to make their service visible and rewardable. We want to turn their daily, invisible labor into a verifiable professional asset that they carry with them. If we try to lock people in, we will fail, just as seventy other countries have

failed. We must build a ladder of opportunity, not a cage of administrative coercion.

Anyone can write a proposal. In this day and age, a machine can easily generate a perfect document in seconds. The rare thing is never the initial idea itself, but rather the person who stays with it through every institutional barrier and slow budget cycle. In the field of implementation research, this dedicated person is called a local champion. The implementation literature repeatedly finds that a local champion matters to whether an intervention survives. I am not able to rank that factor against funding or written guidance, and I should not pretend to a hierarchy the evidence does not give me. Organizational research explains the underlying mechanism as psychological ownership. The concept is straightforward. People invest themselves differently in what they consider to be their own. When an idea is treated as a personal possession, people tend to bring a care and persistence that instruction does not produce. That is an observation about the systems I have worked in, offered as a hypothesis rather than a law.

I lived inside this problem for five years, and this proposal is born from that physical reality. When we needed barcodes for a testing surge, I bought the printing software myself rather than wait for the budget cycle. I mention it for one reason only, and it is not a claim to virtue: buying it was the cheap and slightly embarrassing option, the sort of thing people do in every health system when a process is slower than the work. What it taught me is where such systems actually break, which is never in the part anyone writes a plan about.

What I am proposing is contained and low risk. Hiring a local specialist for a defined term to pilot this registry on our existing legal rails is a cheap option on a massive potential upside. This is a contained experiment with one position, one term, and no new na-

tional budget line. If it succeeds, our public health mission gains a verified history of frontline labor that nobody has ever built before. If it does not succeed, the financial and organizational cost is strictly bounded, and we will document the lessons learned transparently.

However, I must state an essential honesty caveat. Psychological ownership has a well-documented dark side in organizational behavior. The research warns us that having deep psychological ownership over an idea often makes the creator blind to its practical defects. Because of this risk, I might be too close to this project to see its potential flaws. That is why our design must strictly require independent, third-party evaluation. We have to invite outside medical auditors to review the results of our pilot. That commitment is there because I could be wrong about my own design and would rather find out early. Ideas are common, but staying with an idea until it becomes a functional tool for our frontline workforce is what truly matters.

During that time, I learned that the most convincing evidence does not need to be shouted. It only needs to be shown. In the chaotic months of 2021, when our daily testing volume threatened to overwhelm our small team, I did not write a memo requesting a new software procurement. I did not wait for our slow provincial budget cycle to turn. Instead, I paid for the barcoding software myself out of my own pocket and built the first version of our sample-tracking database in a single weekend. That system solved our administrative bottleneck. More importantly, it still runs at our provincial CDC today, five years later, without requiring a single dong of additional government funding. This is not a theoretical proposal or a speculative dream. It is a functional piece of public health infrastructure that has survived the hardest test of all, which is the test of daily use.

Now, we are proposing to expand this concept into a formalized registry to track and reward the continuing medical education and field contributions of our technicians. When we evaluate this next step, we must frame the administrative risk in precise management terms. This is not a wild, unconstrained gamble. Rather, it is a highly contained experiment consisting of exactly one staff position for one defined term. The entire pilot will run on existing legal and regulatory rails, leveraging our current continuing medical education requirements.

If the experiment succeeds, our public health mission gains a powerful asset that nobody in the global health sector has built yet. We will have a verified, portable record of frontline labor that turns invisible daily service into a tangible professional asset. If the experiment fails, the financial and organizational cost is strictly bounded to a single salary. Furthermore, we commit to documenting the entire implementation process and publishing the breakdown transparently so that other provincial programs can learn from our mistakes.

The hardest part of any project is knowing when to let others judge your work. The same attachment that got the barcode database built over a weekend is the attachment that makes me a poor judge of where it fails, and the organizational research on psychological ownership describes both halves of that. Investment brings persistence and adaptive energy. It also brings blindness, because we lose the ability to see our own creations from outside. This is why the proposal requires independent evaluation rather than merely permitting it. (32,34)

This is exactly why the independent evaluation I promised earlier in this proposal is a strict requirement, not a polite courtesy. It is a necessary friction. We must invite outside medical auditors, independent public health experts, and local audit authorities who have

no stake in this system to tear it apart. I would rather find out early that the design is wrong than defend it late. I need independent eyes to verify whether a rotating external quality assessment record can truly prevent data entry fraud without causing staff anxiety. It is the only way to test whether our proposed methods of spot audits and supervisor co-signatures can survive the administrative realities of a busy provincial clinic.

This final note is therefore designed as an open invitation rather than a rigid, top-down implementation plan. I must be modest about what I can see from my laboratory workbench. I do not know your current institutional priorities, your immediate budget constraints, or the political timelines you must navigate. I cannot pretend that a single technician from a provincial laboratory has the full picture of our national health security strategy. I do not know how you weigh educational scholarships against other retention incentives, or how you view our legal continuing medical education rails. But I do know the daily reality of the frontline workforce and the heavy burden of invisible labor. If you look at our preventive medicine network and see a problem here that is genuinely worth solving, I would like to be a part of solving it with you. I would like to know whether this proposed framework fits your vision for our health workers, or if we need to adjust its shape to serve them better.

During those long, exhausting shifts of running diagnostic assays, I learned to separate what we can prove from what we can only guess. One final note, marked clearly as speculation rather than claim. It is a thought that I want to leave on the table, but it should be read as no more than a modest paragraph. In the chaotic days of 2021, a major international donor publicly demanded that a massive emergency relief plan come with accounting transparent enough for the public to see exactly where the money went. In re-

sponse, a detailed plan was produced. It had extensive forecasts, budgets, and spreadsheets. Yet, even after this plan was approved, the money still did not move. I am tempted to read that as evidence that the missing link was never the forecast but the absence of a verifiable record of what had actually been delivered on the ground, by whom, and when.

I believe the same basic design principles we are proposing here would apply wherever frontline public health work is currently invisible. In many remote clinics and laboratories, people are working in the dark, and their efforts are completely unrecorded by central administrators. If we had a simple, decentralized way to verify their daily contributions, we could unlock resources that currently sit idle in capital cities. But I must be completely honest. That is a broad, speculative thought for a later date, and it is absolutely not a claim for now. There is currently no trial evidence, no randomized controlled study, and no peer-reviewed data to support this theory of global health financing. It is my own reflection after years of watching paper-based systems break down under the pressure of a real pandemic.

Therefore, this proposal should be judged only on the very small, concrete thing it actually asks for today. We are not asking to revolutionize international aid or restructure global health security. We are simply asking to fund a single staff position for a twelve-month pilot in one province, running on our existing legal continuing medical education rails to make the invisible labor of our local technicians visible. We want to test whether a simple, secure register can help our staff build a portable professional asset. This is a tiny, low-risk step that requires no new laws or massive budgets. If it fails, we will have learned why cheaply. If it works, we will have shown how to build a ladder of opportunity for those who currently have none. I hope we can start with this small, manageable experiment.

I have watched national health strategies that were beautifully written change nothing at all. Who implements a system matters as much as how it is designed. This truth is best explained through two core ideas that name themselves in the study of organizations.

The first idea is that there is a specific kind of practical knowledge that simply cannot be written down in any manual and can only be acquired by actually doing the work. In our provincial laboratory, we saw this truth play out during the height of the crisis. No national preparedness plan or academic textbook had ever described the administrative and technical steps of linking rapid spreadsheets to a physical barcode printer. Yet, this simple desk task, executed in a back office by someone wearing ordinary clothes before any patient sample even arrived, turned out to be the critical bottleneck for our entire provincial pandemic response. High-level plans can easily describe clinical trials and epidemiological models, but they consistently ignore the silent, mechanical logistics that actually keep a laboratory running.

The second idea comes from a long argument in political economy regarding state capacity and institutional design. Scott's argument in *Seeing Like a State* is that grand schemes fail because the knowledge they need cannot be written down. It lives in the hands of the people doing the work, and it does not survive being turned into a form. Evans and Rauch found in 1999 that states which recruit on merit and offer predictable careers tend to grow faster. Which way the causation runs is still argued over, and I am not going to pretend a study of national bureaucracies settles anything about one provincial laboratory. Top-down planners often assume that a standardized blueprint can be handed to anyone to produce the same result, but this assumption ignores the unique context of each clinic (72).

This proposal is an attempt to write down the small part of the system that can actually be written down in software and regulations, written by someone who learned the unwritten rest of the reality the only way it can be learned: by living it. I did not learn any of this from a book, and the success of the design rests on the technicians who would have to carry it, not on the person who drew it.

The machine records when it ran, which thermal programme was used, who was logged in, and what the raw result was. This telemetry is synchronized directly to the cloud without passing through anyone who could edit it. That is captured where it is produced, and it is the one part of the proposal nobody can quietly adjust afterwards. I should say plainly what it is not. The instrument knows that a run began at a given minute under a given login. It does not know whether the master mix was prepared correctly, whether the specimen was the one on the form, or whether the person logged in was the person at the bench. Telemetry fixes the time and the account. Everything about competence is established elsewhere, or not at all.

The question of whether artificial intelligence should monitor or even operate such a system is real, but it belongs strictly to a later phase. A first pilot must be small enough that failing is cheap. This is a contained experiment. If it works, the mission gains something nobody has built yet.

The honest limit here is this. This remains untested, as stated at the outset. We are proposing a logical, evidence-informed design to test these ideas. By starting with one machine and one connection, we can prove our concept without risking scarce resources on a sprawling failure.

I want to be exact about what already exists and what is genuinely missing, so that nobody can accuse us of reinventing the wheel. Instrument telemetry that reports performance data back to inter-

national funders is not a new idea. In fact, one major connectivity platform already collects and aggregates data from molecular testing devices across forty-three countries. This platform automatically tracks raw results, error rates, reagent lot numbers, calibration dates, and instrument serial numbers. It is a highly sophisticated data stream.

However, that existing system has a massive, critical blind spot. It attributes all of this testing activity to an inanimate serial number and a physical facility site, rather than to a named operator who actually performed the work. Furthermore, it can only see instruments that have been successfully installed and connected to the network. This means that the worst failures of international donation programs are completely invisible to it. The literature itself notes that the vast pools of data collected by these platforms remain largely untapped.

These documented failures of donated equipment are real and specific, rather than rhetorical. For example, one study in Indonesia found that twenty-five out of twenty-six district laboratories owned polymerase chain reaction machines, but only a single laboratory was still running the testing service. Similarly, a rigorous assessment in Madagascar found that fewer than a quarter of donated devices were actively performing tests, while more than half of the equipment had never been installed at all.

However, we must remain entirely honest about the statistics we repeat to justify our funding. The widely circulated claim that seventy to ninety-six percent of donated medical equipment in developing countries never works is simply not evidentially grounded. The actual, defensible measured figure is thirty-eight percent of equipment out of service, calculated across a massive dataset of one hundred and twelve thousand individual items. Furthermore, we must state equally honestly that no comparable, systematic study of

equipment downtime exists for Vietnam. Therefore, we cannot assert anything about the operational status of Vietnamese laboratories without dedicated, structured fieldwork.

This proposal does not ask anyone to spend millions of dollars building new telemetry hardware or duplicating existing networks. Instead, we are asking to add the single missing layer of human accountability to an infrastructure that is already physically running on the ground. We want to connect an existing machine serial number to a real technician's name, and then connect that technician's name to a life-changing professional opportunity. By starting with this simple link, we can turn a cold serial number into a portable, verifiable professional asset for our frontline workers.

A beautiful analysis is useless if the raw data is broken. Today, we live in an age of incredibly powerful computational models. The reasoning tools available to us now are far beyond what existed in 2019. Yet, the data underneath them has not changed. No model, no matter how advanced, can read what nobody bothered to write down in the first place. The real bottleneck of health security sits at the source of the record rather than at the level of analysis. Adding a stronger model on top of a weak foundation does not repair the foundation itself. (73)

This structural gap explains an honest admission made earlier in our planning. Both the World Health Organization and the Cochrane reviews attach low or very low certainty to almost every recommendation regarding health worker retention and performance in remote areas. This is not due to a shortage of brilliant analysts or computers in Geneva or London. It is a direct consequence of a weakness in the underlying records themselves. The studies are poorly characterized, lacking machine-enforced audit trails and individual attribution. We cannot build a high-certainty science of

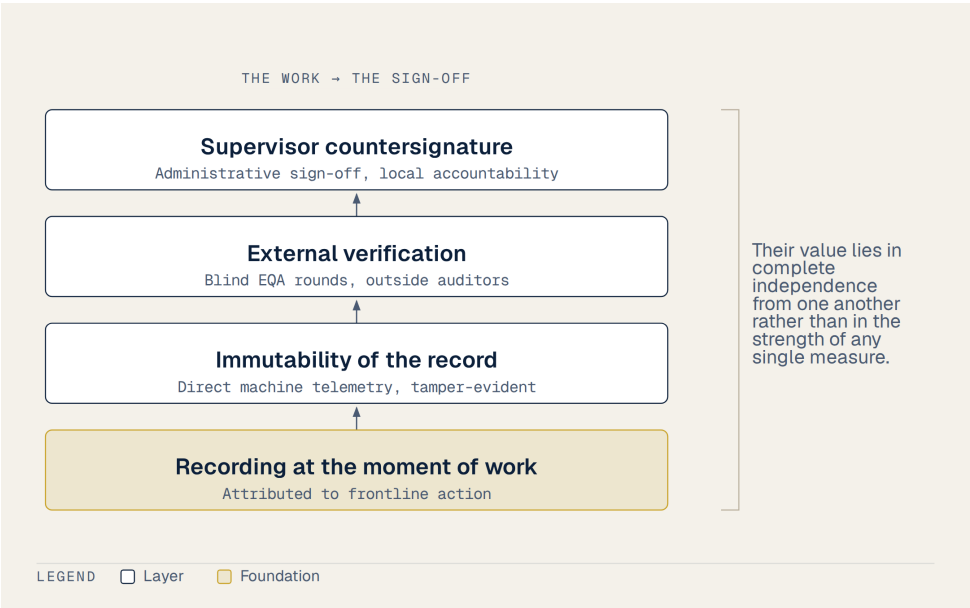
public health on top of a pile of vague, unverified administrative summaries. (16,74)

Yet, there is a deeply optimistic half to this argument. Information that nobody bothered to record today may carry a value that we cannot possibly guess at the moment of recording. A sample run that seems routine on a quiet Tuesday afternoon might hold the key to tracking a mutated pathogen years from now. If we capture a record that is attributable, contemporaneous, and externally verified, we create a durable asset. It becomes something that can still be read and mined for insights years later, rather than a piece of administrative paperwork designed only to be filed away and forgotten.

We need to state clearly what we are not claiming. We do not promise that any artificial intelligence model will magically detect the next outbreak on its own. We promise no particular technology, nor do we claim that machine learning will solve the human crises of our provincial clinics. The question of whether software should monitor or operate our laboratory instruments belongs strictly to a later phase of this work. A first pilot must remain deliberately small and focused on the simple act of recording. I have to first ensure that when a technician does the work, a secure, immutable record of that work actually exists. The same caveat applies to the point-accumulation model. We are proposing a logical, evidence-informed design to test these behavioral ideas. By securing the recording layer first, we can build a foundation that is worthy of the powerful models waiting above it.

The most dangerous illusions are the ones that promise total safety. In writing this proposal, I want to make a final, blunt statement of position. I want to place it clearly so that no reader can mistake this book or this system for something it is not. This is not a cure. This is not a shield. There is no single measure that makes

the next pandemic something we no longer need to worry about. Anyone offering one should be read with suspicion.



The value is in the independence, not the strength. Four barriers that fail in different ways. A system whose layers all fail the same way has one layer.

What I am describing is one hole in one layer of the defences. It is the Swiss cheese model of accidents, where every layer of defence has holes in it, where a single error usually passes through without causing any harm, and where disaster happens only when the holes in successive layers happen to line up. The most important thing this model says is not that somebody made a mistake. Instead, it shows how everyone involved can believe they are doing the right thing while the system still carries them toward a fatal outcome. This is the exact trap we see when an entire peer group drifts together, making the accurate laboratory look like the one that is wrong.

The corresponding discipline from industrial safety is to isolate a hazard and then verify that isolation rather than assume it. This is the same instinct as sending a blind sample to a laboratory instead of asking it how it is doing. What follows for this proposal is that no single layer is being offered as sufficient. Recording at the moment of work is one layer. Immutability of that record is a second. External verification is a third. A supervisor countersignature is a fourth. The value lies in these layers being completely independent of one another rather than in any of them being exceptionally strong.

This is where the evidence stops. It is untested, like everything else here. The World Health Organization and Cochrane reviews attach low or very low certainty to almost every recommendation in this field. We do not claim that any model will detect the next outbreak on its own. This proposal promises no particular technology, and the question of whether software should monitor or operate instruments belongs strictly to a later phase. A proposal that knows its own size can be tested modestly, funded easily, and abandoned cheaply if it fails. We refuse to sell a grand illusion when a simple, honest patch is what actually protects us. (27,34)

We tend to treat global health security networks as permanent structures. We assume that once a system is built, it remains in place. But that is an illusion. The arrangements that look permanent to us are usually maintained day by day rather than settled once and for all. Peace, stability, and institutional readiness are not monuments. They are activities.

Institutions built after a catastrophe to prevent the next one rarely fail because someone destroys them. They fail because the people who inherit them stop paying for maintenance, and the decay is quiet enough that nobody has to decide to allow it. The Korean armistice of 1953 holds seven decades of quiet not because it settled anything, but because people keep choosing to honour a docu-

ment meant to be temporary; the League of Nations was hollowed out by members who declined to act when acting grew expensive. I raise the comparison briefly and then leave it, because a laboratory is not a treaty and I do not want to argue from analogy any further than it carries. (74,75)

We must not claim that history could have gone otherwise. Counterfactual history is not evidence, and our argument does not need it. The point is much narrower, and it is more than enough for our purposes. Capability is a flow rather than a stock. It is a river that must be fed, not a pool of water that sits in a tank. When the funding, the staffing, and the collective attention stop, the capability dries up.

The small proposal in this book is one attempt to keep a very small part of that flow running. We do not promise a grand historical solution, nor do we offer a permanent shield against future crises. We are asking only to build a modest, attributable, and verified record of daily laboratory work. The pilot exists precisely because none of this has been measured. We are proposing a logical, evidence-informed twelve-month pilot. By connecting a machine's serial number to a technician's name, we can feed the human capability of our frontline laboratories, so that a small part of the global defence system does not quietly fade into neglect.

What this looks like for one person

Everything above is machinery. Machinery is not what persuades anyone, so let me describe three people walking through it. None of them exists. These are illustrations of how the mechanism is meant to work, not case studies, and I want that stated plainly rather than buried in a footnote. Nobody has run this yet. That is what a pilot is for.

The first is a laboratory technician at a district health centre. She has no publications. She has never presented at a conference. Her English is enough to read a package insert and not much more. On paper she has nothing to submit anywhere, and if you asked her supervisor for a reference letter he would write that she is hardworking and reliable, which is true and worth nothing, because every letter says that. After eighteen months on a verified record she has something else. The record shows which assays she ran and how many. It shows that she sat three rounds of external quality assessment and passed them. Her letter can now carry a number instead of an adjective, and the number was checked by someone outside her own institution. What she was short of was never competence. It was evidence.

The second collects samples at commune level. Her record shows an unbroken chain of custody and, over many months, not one specimen rejected for a labelling or handling fault. She applies for a position at the provincial centre. I want to be careful here, because this is the point where an honest reader objects. A verified record does not abolish the advantage of knowing the right people, and I am not going to pretend that it does. What it does is narrower. It gives someone without that advantage one thing to put on the table that can be checked, where previously she had nothing at all. That is a smaller claim than fairness. It is the one I can defend.

The third has already left. She worked nine years in a province that no longer exists, merged into another in June 2025. There is no office to write to, no personnel department still holding her file under a name that still appears on a map. Under the arrangement I am describing, the record was never held by the office. It was held by her, countersigned by the quality assessment scheme that verified it, and it does not care that the letterhead has changed. She can still show what she did. This is the case I find hardest to argue

against and easiest to forget, because the people it describes are, by definition, already gone. (21)

None of the three is promised a career. The record does not hire anyone, does not fund anyone, and cannot make an institution care. It does one thing. It converts work that was real but invisible into something a stranger can check. What anyone does with that afterwards is not mine to promise.

There is one more thing to say, and it is the reason I wrote any of this.

Every pandemic plan ever written contains an assumption that appears in no budget line. The assumption is that when the moment comes, people will do more than they are paid to do. They will stay for the extra shift. They will live away from their families for months. They will accept a level of risk that no employment contract obliges them to accept. Plans call this surge capacity, as though it were a quantity of equipment held in a warehouse. It is not. It is a decision that several thousand individual people make, each one on their own, usually within the first week.

That willingness is a real resource, and it was spent in 2021. What matters is what happened afterwards. Nine thousand six hundred and eighty people left the public system in the eighteen months that followed. The occupational allowance was raised and then, on the first of January 2024, returned to the level set in 2011. The preventive share of the health budget continued its slow crawl toward a target set in 2008. None of these is a scandal. Taken together they are something more consequential: a promise made under pressure and quietly unmade once the pressure lifted. (5,6,19)

People remember that. I remember it, and so do the colleagues I worked beside. I will be plain, because I do not think anything is gained by being delicate about it. I do not believe we would give what we gave in 2021 a second time. That is not a failure of charac-

ter in the health workforce. It is an accurate reading of the available evidence, made by people who are trained to read evidence.

This is what makes the argument in this book something other than a request for kindness. A country cannot appropriate its way to a surge. The surge happens only if the people who would have to carry it believe that this time will differ from last time, and belief of that kind is built from records, not from speeches. If a system cannot show what a person did during the last emergency, it has nothing to point to when it asks them to do it again. Recognition is not a reward handed out after the work is finished. It is the infrastructure that makes the next mobilisation possible.

That is the hole I am trying to patch. It is one hole. There are others, and better people are working on them. But this one is mine, I have stood in it, and I know exactly how deep it goes.

(5,6,19,74) # References {.nochap}

1. Dieleman M, Cuong PV, Anh LV, Martineau T. Identifying factors for job motivation of rural health workers in north viet nam. *Human Resources for Health*. 2003;1:10.
2. Tran BX, Van Hoang M, Nguyen HD. Factors associated with job satisfaction among commune health workers in vietnam. *Global Health Action*. 2013.
3. Cutler DM, Summers LH. The COVID-19 pandemic and the \$16 trillion virus. *JAMA*. 2020;324(15):1495–6. doi:[10.1001/jama.2020.19759](https://doi.org/10.1001/jama.2020.19759)
4. G20 High Level Independent Panel on Financing the Global Commons for Pandemic Preparedness and Response. A global deal for our pandemic age. G20; 2021.
5. Bộ Y tế Việt Nam. Báo cáo về số lượng nhân viên y tế thôi việc, bỏ việc, 01/01/2021 – 30/6/2022. <https://suckhoedoisong.vn/18-thang-co-9680-nhan-vien-y-te-xin-thoi-viec-bo-viec-bo-y-te-chi-ra-4-nguyen-nhan-chinh-169220918125927961.htm>; 2022.
6. Government of Viet Nam. Nghị định 05/2023/NĐ-CP: Sửa đổi mức phụ cấp ưu đãi nghề đối với viên chức y tế dự phòng và y tế cơ sở. 2023.
7. Grobler L, Marais BJ, Mabunda S. Interventions for increasing the proportion of health professionals practising in rural and other underserved areas. *Cochrane Database of Systematic Reviews*. 2015.
- 8.

- Diaconu K, Falconer J, Verbel A, Fretheim A, Witter S. Paying for performance to improve the delivery of health interventions in low- and middle-income countries. *Cochrane Database of Systematic Reviews*. 2021.
9. World Health Organization. WHO guideline on health workforce development, attraction, recruitment and retention in rural and remote areas. Geneva: WHO; 2021.
 10. Miyazaki M et al. Motivations of rural health workers in lao PDR: A discrete choice experiment. *Human Resources for Health*. 2013;11:22.
 11. Rockers PC et al. Preferences for working in rural clinics among trainee health professionals in timor-leste: A discrete choice experiment. *PLOS ONE*. 2016;11(11):e0165940.
 12. World Health Organization. Increasing access to health workers in remote and rural areas through improved retention: Global policy recommendations. Geneva: WHO; 2010.
 13. Willis-Shattuck M, Bidwell P, Thomas S, Wyness L, Blaauw D, Ditlopo P. Motivation and retention of health workers in developing countries: A systematic review. *BMC Health Services Research*. 2008;8:247.
 14. Turcotte-Tremblay AM, Gali-Gali IA, Ridde V. An exploration of the unintended consequences of performance-based financing in 6 primary healthcare facilities in burkina faso. *International Journal of Health Policy and Management*. 2022.
 15. Marks IH, Thomas H, Bakhet M, Fitzgerald E. Medical equipment donation in low-resource settings: A review of the literature and guidelines for surgery and anaesthesia in low-income and middle-income countries. *International Health*. 2019;11(5):379–86.
 16. World Health Organization and World Bank. Analysis of pandemic preparedness and response (PPR) architecture, financing needs, gaps and mechanisms. Report to the G20 Joint Finance; Health Task Force; 2022.
 17. Quốc hội nước Cộng hoà xã hội chủ nghĩa Việt Nam. Nghị quyết số 18/2008/QH12 về đẩy mạnh thực hiện chính sách, pháp luật xã hội hoá để nâng cao chất lượng chăm sóc sức khoẻ nhân dân. 2008.
 18. Quốc hội nước Cộng hoà xã hội chủ nghĩa Việt Nam. Nghị quyết số 99/2023/QH15 giám sát chuyên đề về việc huy động, quản lý và sử dụng các nguồn lực phục vụ công tác phòng, chống dịch COVID-19; việc thực hiện chính sách, pháp luật về y tế cơ sở, y tế dự phòng. 2023.
 19. Cổng Thông tin điện tử Quốc hội nước Cộng hoà xã hội chủ nghĩa Việt Nam. Ngân sách y tế cho y tế dự phòng ở một số địa phương chưa đạt mục tiêu quốc hội giao. <https://quochoi.vn/tintuc/Pages/tin-hoat-dong-cua-quoc-hoi.aspx?ItemID=74480>; 2023.
 - 20.

- Đoàn giám sát của Quốc hội khoá XV. Báo cáo số 455/BC-ĐGS ngày 19/5/2023 về kết quả giám sát chuyên đề "việc huy động, quản lý và sử dụng các nguồn lực phục vụ công tác phòng, chống dịch COVID-19; việc thực hiện chính sách, pháp luật về y tế cơ sở, y tế dự phòng". Quốc hội Việt Nam; 2023 May.
21. Quốc hội nước Cộng hoà xã hội chủ nghĩa Việt Nam. Nghị quyết số 202/2025/QH15 về việc sắp xếp đơn vị hành chính cấp tỉnh. <https://xaydungchinhhsach.chinhphu.vn/toan-van-ngghi-quyet-so-202-2025-qh15-ve-sap-xep-don-vi-hanh-chinh-cap-tinh-119250612174148722.htm>; 2025.
 22. 1EdTech Consortium. Open badges specification version 3.0. 2024.
 23. World Health Organization. Global digital health certification network (GDHCN) and SMART trust. 2024.
 24. Harkin B, Webb TL, Chang BPI, Prestwich A, Conner M, Kellar I, et al. Does monitoring goal progress promote goal attainment? A meta-analysis of the experimental evidence. *Psychological Bulletin*. 2016;142(2):198–229.
 25. Kurtessis JN, Eisenberger R, Ford MT, Buffardi LC, Stewart KA, Adis CS. Perceived organizational support: A meta-analytic evaluation of organizational support theory. *Journal of Management*. 2017;43(6): 1854–84.
 26. Rousseau DM. *Psychological contracts in organizations*. Sage; 1995.
 27. Sailer M, Homner L. The gamification of learning: A meta-analysis. *Educational Psychology Review*. 2020;32:77–112.
 28. World Health Organization. Guideline on data integrity. WHO Technical Report Series 1033, Annex 4; 2021.
 29. Easterly W, Levine R, Roodman D. Aid, policies, and growth: comment. *American Economic Review*. 2004;94(3):774–80.
 30. Nunes JC, Drèze X. The endowed progress effect: How artificial advancement increases effort. *Journal of Consumer Research*. 2006;32(4):504–12.
 31. Kivetz R, Urminsky O, Zheng Y. The goal-gradient hypothesis resurrected: Purchase acceleration, illusory goal progress, and customer retention. *Journal of Marketing Research*. 2006;43(1):39–58.
 32. Deci EL, Koestner R, Ryan RM. A meta-analytic review of experiments examining the effects of extrinsic rewards on intrinsic motivation. *Psychological Bulletin*. 1999;125(6):627–68.
 33. Gerhart B, Fang M. Pay, intrinsic motivation, extrinsic motivation, performance, and creativity in the workplace. *Annual Review of Organizational Psychology and Organizational Behavior*. 2015;2:489–521.

34. Cerasoli CP, Nicklin JM, Ford MT. Intrinsic motivation and extrinsic incentives jointly predict performance: A 40-year meta-analysis. *Psychological Bulletin*. 2014;140(4):980-1008.
35. Colquitt JA, Conlon DE, Wesson MJ, Porter COLH, Ng KY. Justice at the millennium: A meta-analytic review of 25 years of organizational justice research. *Journal of Applied Psychology*. 2001;86(3):425-45.
36. Van den Broeck A, Ferris DL, Chang CH, Rosen CC. A review of self-determination theory's basic psychological needs at work. *Journal of Management*. 2016;42(5):1195-229.
37. Deci EL, Olafsen AH, Ryan RM. Self-determination theory in work organizations: The state of a science. *Annual Review of Organizational Psychology and Organizational Behavior*. 2017;4:19-43.
38. Kerfoot BP, Baker H, et al. An online spaced-education game among clinicians improves their patients' time to blood pressure control: A randomized controlled trial. *Circulation: Cardiovascular Quality and Outcomes*. 2014.
39. Gentry SV, Gauthier A, et al. Serious gaming and gamification education in health professions: Systematic review. *Journal of Medical Internet Research*. 2019;21(3):e12994.
40. Kok MC, Dieleman M, Taegtmeyer M, et al. Which intervention design factors influence performance of community health workers in low- and middle-income countries? A systematic review. *Health Policy and Planning*. 2015;30(9):1207-27.
41. OpenLMIS Community. OpenLMIS: Qualified logistics management information system software solution. 2019.
42. FIND. A guide to diagnostic network optimization. FIND; 2022.
43. Jones KE, Patel NG, Levy MA, Storeygard A, Balk D, Gittleman JL, et al. Global trends in emerging infectious diseases. *Nature*. 2008;451:990-3.
44. Allen T, Murray KA, Zambrana-Torrel C, et al. Global hotspots and correlates of emerging zoonotic diseases. *Nature Communications*. 2017;8:1124.
45. Rulli MC, D'Odorico P, Galli N, Hayman DTS. Land-use change and the livestock revolution increase the risk of zoonotic coronavirus transmission from rhinolophid bats. *Nature Food*. 2021;2:409-16.
46. United States Government. U.s. Government global health security strategy 2024 [Internet]. 2024. Available from: <https://bidenwhitehouse.archives.gov/wp-content/uploads/2024/04/Global-Health-Security-Strategy-2024-1.pdf>
47. United States Department of State. America first global health strategy. <https://www.state.gov/releases/office-of-the-spokesperson/2025/09/release-of-the-america-first-global-health-strategy>; 2025.

48. Frieden TR, Lee CT, Bochner AF, Buissonnière M, McClelland A. 7-1-7: An organising principle, target, and accountability metric to make the world safer from pandemics. *The Lancet*. 2021;398(10300):638–40.
49. NHS England, Medical Specialty Training. Core surgical training: Portfolio guidance for candidates (intercollegiate surgical eLogbook consolidation report). <https://medical.hee.nhs.uk/medical-training-recruitment/medical-specialty-training/surgery/core-surgery/core-surgical-training-portfolio-guidance-for-candidates>; 2025.
50. US Federal Aviation Administration. 14 CFR 61.159, aeronautical experience: Airplane category rating; 14 CFR 61.160, restricted privileges; 14 CFR 121.436, pilot qualification. <https://www.ecfr.gov/current/title-14/chapter-I/subchapter-D/part-61/subpart-G>; 2013.
51. National Board for Professional Teaching Standards. National board certification: Portfolio requirements and state financial incentives. https://www.nbpts.org/wp-content/uploads/2021/05/state_incentive_chart.pdf; 2025.
52. Landgraf KM, Kakkar R, Meigs M, et al. Open-source laboratory information systems in vietnam: The path toward sustainability and host country ownership. *International Journal of Medical Informatics*. 2016.
53. Ministry of Health, Viet Nam. Quyết định 2429/QĐ-BYT: Tiêu chí đánh giá mức chất lượng phòng xét nghiệm y học. 2017.
54. US Centers for Medicare and Medicaid Services. 42 CFR §493.801: Condition – enrollment and testing of samples (proficiency testing). 2024.
55. Van HT, Tran VT, Ha MT, Vu QH. Model of implementing proficiency testing in vietnam, a developing country. *Practical Laboratory Medicine*. 2023.
56. Nguyen VTT, Best S, Pham HT, Troung TXL, Hoang TTH, Wilson K, et al. HIV point of care diagnosis: Preventing misdiagnosis experience from a pilot of rapid test algorithm implementation in selected communes in vietnam. *Journal of the International AIDS Society*. 2017;20(Suppl 6): 21752. doi:[10.7448/IAS.20.7.21752](https://doi.org/10.7448/IAS.20.7.21752)
57. United Kingdom Accreditation Service. Assessment of a medical laboratory's approach to the assurance of clinical staff competence and use of EQA. *UKAS Technical Bulletin*; 2017.
58. American Society for Clinical Pathology, Board of Certification. Medical laboratory scientist, MLS(ASCP): Eligibility routes. <https://www.ascp.org/boc/explore-credentials/view-all-credentials/MLS>; 2026.
59. Batalova J, Fix M. Leaving money on the table: The persistence of brain waste among college-educated immigrants. *Migration Policy Institute*; 2021.
60. Adams C, Cain P, Pinkas D, Zuccherato R. Internet x.509 public key infrastructure time-stamp protocol (TSP). 2001.

61. Ministry of Health, Viet Nam. Thông tư 32/2023/TT-BYT: Quy định chi tiết một số điều của luật khám bệnh, chữa bệnh. 2023.
62. ORCID. ORCID: Connecting research and researchers. 2024.
63. European Data Protection Board. Guidelines 02/2025 on processing of personal data through blockchain technologies. 2026.
64. Akerlof GA. The market for "lemons": Quality uncertainty and the market mechanism. *Quarterly Journal of Economics*. 1970;84(3):488-500.
65. Spence M. Job market signaling. *Quarterly Journal of Economics*. 1973;87(3):355-74.
66. Ethiopia Ministry of Health and JSI. The feasibility and effectiveness of digital-enabled performance-based incentives in ethiopia's community health program. 2025.
67. Wagner T, Hennig-Thurau T, Rudolph T. Does customer demotion jeopardize loyalty? *Journal of Marketing*. 2009;73(3):69-85.
68. Boysen PG. Just culture: A foundation for balanced accountability and patient safety. *Ochsner Journal*. 2013;13(3):400-6.
69. Bevan G, Hood C. What's measured is what matters: Targets and gaming in the english public health care system. *Public Administration*. 2006;84(3):517-38.
70. US Health Resources and Services Administration. National health service corps scholarship program: Report to congress. 2021.
71. Frehywot S, Mullan F, Payne PW, Ross H. Compulsory service programmes for recruiting health workers in remote and rural areas: Do they work? *Bulletin of the World Health Organization*. 2010;88(5):364-70.
72. Evans P, Rauch JE. Bureaucracy and growth: A cross-national analysis of the effects of "weberian" state structures on economic growth. *American Sociological Review*. 1999;64(5):748-65.
73. US Food and Drug Administration. 21 CFR part 11: Electronic records; electronic signatures. 1997.
74. Global Preparedness Monitoring Board. A world at risk: Annual report on global preparedness for health emergencies. WHO; World Bank; 2019.
75. United Nations Command and Korean People's Army and Chinese People's Volunteers. Korean armistice agreement. 1953.